Multi-Mode RF Communications Subsystem for a 3U CubeSat

Flowgraph Configuration and Signal Validation Report

NIRO LAB, Communications Subsystem

1. Introduction

The configuration and functional validation of a multi-mode radio frequency communications subsystem created for a 3U CubeSat platform are described in this report. Routine command and telemetry, dual-channel and high-rate payload downlink, an inter-satellite link, and an emergency low-frequency return path are among the five operational modes that the subsystem supports. A pair of GNU Radio Companion flowgraphs, one for the ground station (called GND) and one for the satellite terminal (called STL), are used to realize each mode.

Each flowgraph was run to record baseband signals, which were then processed by an offline Python visualization pipeline to produce the two- and three-dimensional figures shown here. Additionally, for every mode, the live GNU Radio runtime plots generated by each flowgraph are included. The document is arranged according to mode. The signal chain is explained, the recovered signal figures and runtime plots are displayed, and the satellite and ground flowgraph canvases are displayed for each mode. Telemetry is received as a downlink, and telecommand is generated as an uplink on the ground segment because the transmit and receive roles are switched with respect to the satellite.

2. Hardware Configuration

The communications subsystem is designed around a sophisticated software-defined radio transceiver coupled with a set of deployable antennas, ensuring versatility and efficiency in communication. At the heart of this subsystem is the primary transceiver, a GomSpace NanoCom class module that utilizes the advanced AD9361 radio front end integrated with a Zynq system on chip. This innovative front end is engineered to support multi-band operations, enabling the realization of all five operational modes from a unified hardware platform.

For the UHF and VHF communication links, a crossed dipole turnstile antenna is employed. This antenna configuration is specifically chosen for its ability to provide near omnidirectional coverage, which is crucial for effective command, telemetry, and beacon functions. In contrast, the S-band payload links are facilitated by a dedicated patch antenna, optimized for high-frequency communication.

The subsystem's design emphasizes modularity and adaptability, allowing for seamless transitions between different operational modes. The three main hardware components, which are integral to the functionality of the communications subsystem, are illustrated in the accompanying diagrams below. This detailed configuration ensures robust performance across various communication scenarios, enhancing the overall capabilities of the CubeSat platform.

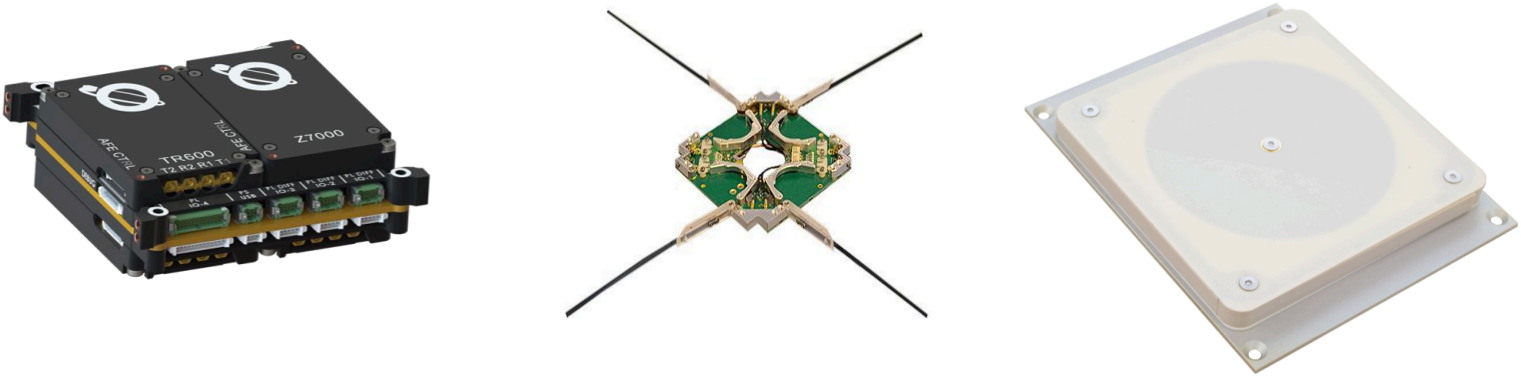


Figure 2. Hardware components. Left: GomSpace NanoCom TR600 transceiver with the Zynq SoC board. Center: UHF and VHF turnstile antenna. Right: S-band patch antenna.

The high-level architecture is summarized in the system block diagram below. All five operational modes are realized from a single shared SDR transceiver, which routes command, telemetry, and beacon traffic through the UHF and VHF turnstile antenna and payload data through the S-band patch antenna. The ground segment mirrors the satellite with its own antenna and SDR.

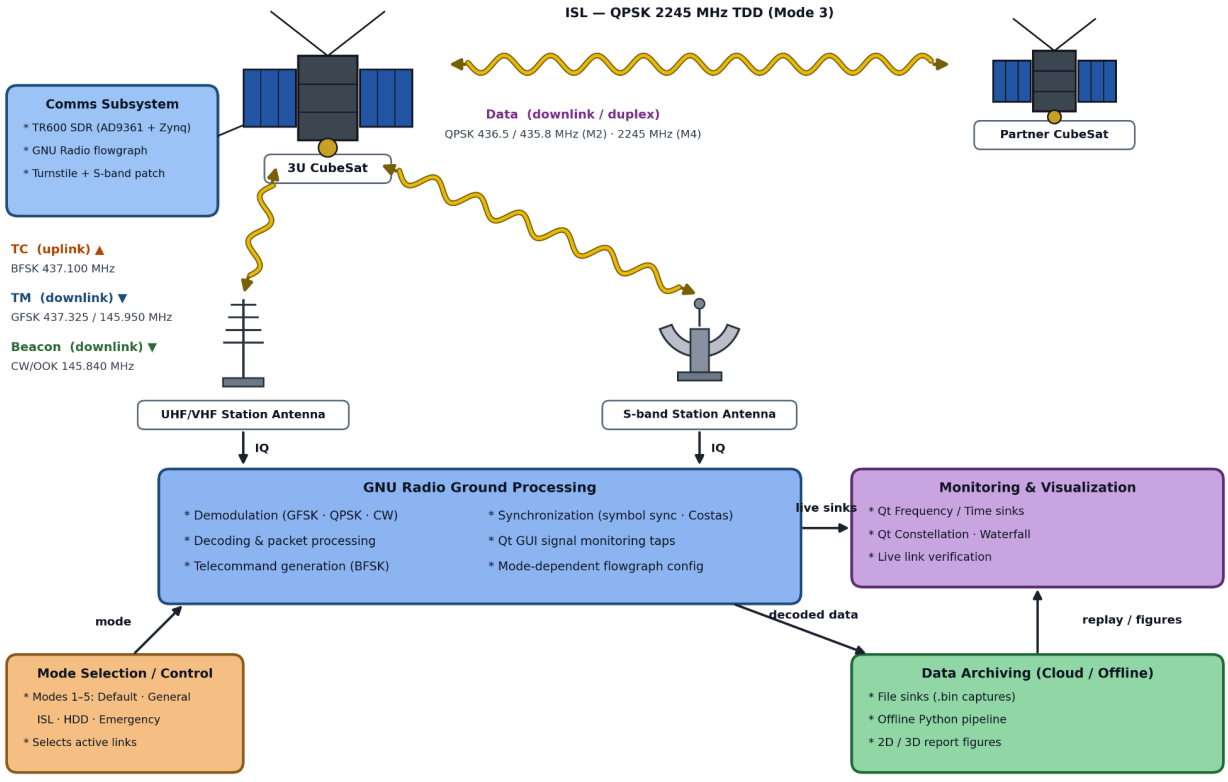


Figure 3. System block diagram. Five operational modes are correlated with the DR transceiver, which connects to the UHF/VHF turnstile and S-band patch antennas with RF links to the ground segment.

3. Operational Modes Overview

The signal assignments for the five operational modes are summarized in the two tables below. The first table gives the satellite terminal view, and the second gives the ground station view. Because the transmit and receive roles are reversed between the two segments, the telecommand and telemetry columns swap direction between the tables, while the payload data and beacon assignments follow the mode definition.

Satellite Communication Modes

Mode	Telecommand (BFSK) *437.100 MHz*	Telemetry (GFSK)	Data (QPSK)	VHF Beacon Signal (CW/OOK)
Mode-1 (Default)	UHF-RX *437.100 MHz*	UHF-TX *437.325 MHz*	0	0
Mode-2 (General)	UHF-RX *437.100 MHz*	UHF-TX *437.325 MHz*	UHF-TXRX *436.500 / 435.800 MHz*	Beacon *145.84 MHz*
Mode-3 (ISL)	UHF-RX *437.100 MHz*	UHF-TX *437.325 MHz*	S-TXRX (ISL) *2245.000 MHz*	Beacon *145.84 MHz*
Mode -4 (HDD)	UHF-RX *437.100 MHz*	UHF-TX *437.325 MHz*	S-TX *2245.000 MHz*	Beacon *145.84 MHz*
Mode - 5 (Emergency)	UHF-RX *437.100 MHz*	VHF-TX *145.950 MHz*	0	Beacon *145.84 MHz*

Ground Station Communication Modes

Mode	Telecommand (BFSK)	Telemetry (GFSK)	DC (QPSK)	BEACON (CW/OOK) *145.84 MHz*
Mode-1 (Default)	UHF-TX *437.100 MHz*	UHF-RX *437.325 MHz*	0	0
Mode 2 (General)	UHF-TX *437.100 MHz*	UHF-RX *437.325 MHz*	UHF-TXRX *436.500/435.800 MHz*	VHF RX *145.84 MHz*
Mode 3 (ISL)	UHF-TX *437.100 MHz*	UHF-RX *437.325 MHz*	0	VHF RX *145.84 MHz*
Mode -4 (HDD)	UHF-TX *437.100 MHz*	UHF-RX *437.325 MHz*	S-RX *2245.000 MHz*	VHF RX *145.84 MHz*
Mode - 5 (Emergency)	UHF-TX *437.100 MHz*	VHF-RX *145.950 MHz*	0	VHF RX *145.84 MHz*

All modes share a common telecommand uplink using binary frequency shift keying at 437.100 MHz with a deviation of 3000 Hz. Telemetry uses Gaussian frequency shift keying at 437.325 MHz in Modes 1 through 4 and moves to 145.950 MHz in the emergency mode. The beacon, where present, is a continuous wave carrier at 145.840 MHz keyed at 8.333 Hz, which corresponds to 20 words per minute. Payload data uses quadrature

phase shift keying in a dual UHF frequency division arrangement in Mode 2 and in the S-band at 2245 MHz for the inter-satellite and high-data modes.

4. Mode-by-Mode Analysis

4.1 Mode 1: Default / UHF

Satellite Terminal (STL)

Mode 1 is the baseline command and telemetry configuration with no payload data channel and no beacon. On the satellite, the telecommand receiver path decodes the uplink binary frequency shift keying signal centered at 437.100 MHz. A source vector is repacked, mapped through chunks to symbols, repeated with an interpolation of 40, and frequency-modulated at a sample rate of 48 kHz. Recovery is performed through a channel model, a low-pass filter, a Mueller and Muller symbol synchronizer, a binary slicer, and a correlated access code block that tags each frame using access code 101010 before repacking to bytes and writing decoded_tc.bin.

The telemetry transmit path uses Gaussian frequency shift keying at 437.325 MHz with 5 samples per symbol, sensitivity of 500 m, and a BT product of 0.35. The modulated stream passes through a channel model and is demodulated by a GFSK demodulator to confirm recovery of the telemetry bit stream.

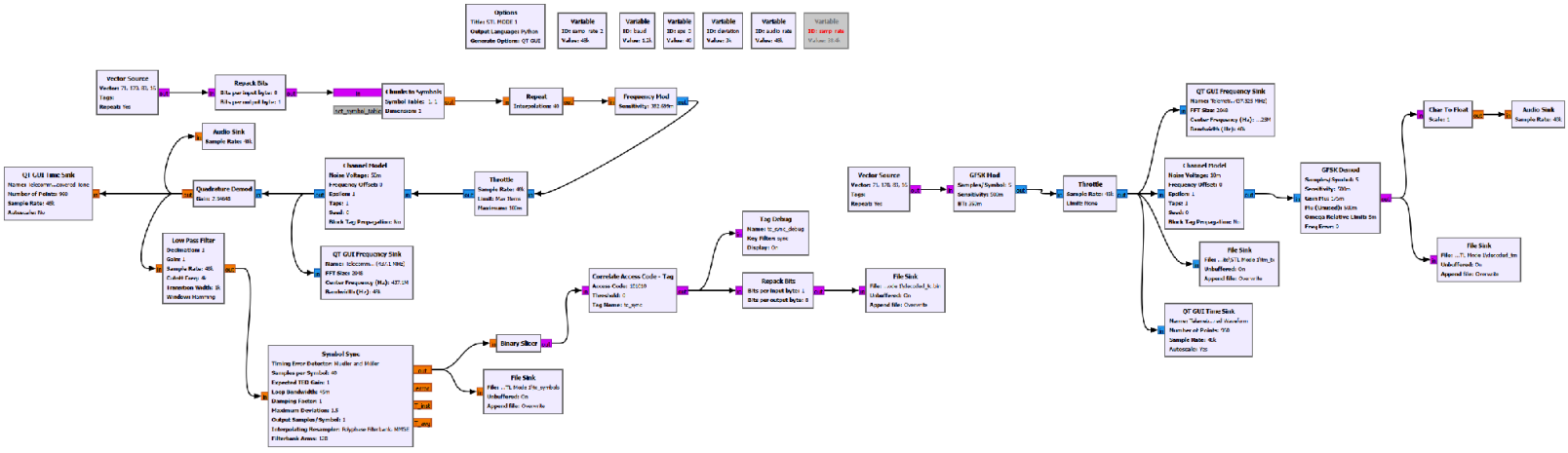


Fig. 1.1 Satellite (STL) flowgraph canvas.

GNU Radio Runtime Plots (STL)

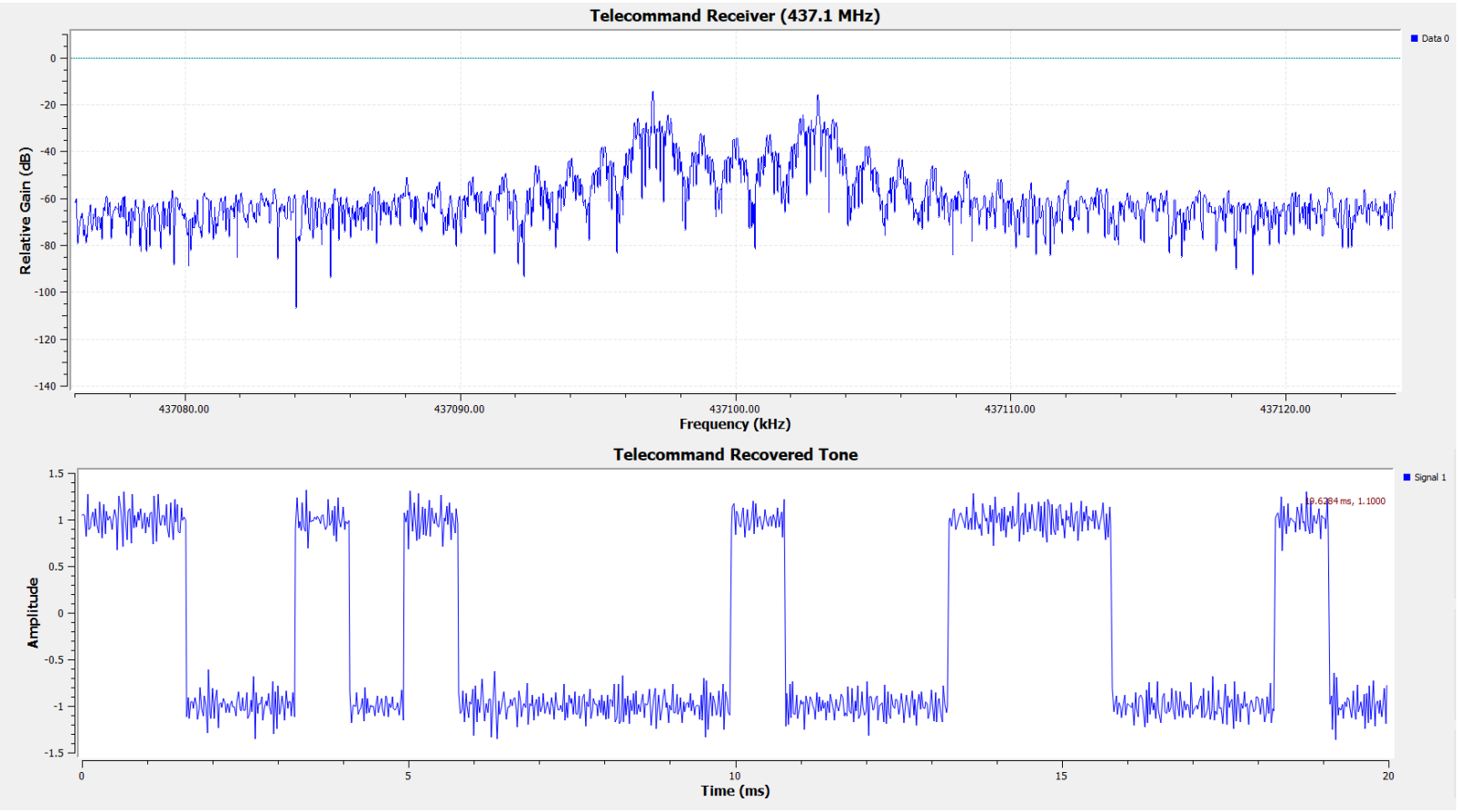


Fig. 1.2 (STL) Telecommand receiver spectrum and recovered tone (437.1 MHz BFSK)

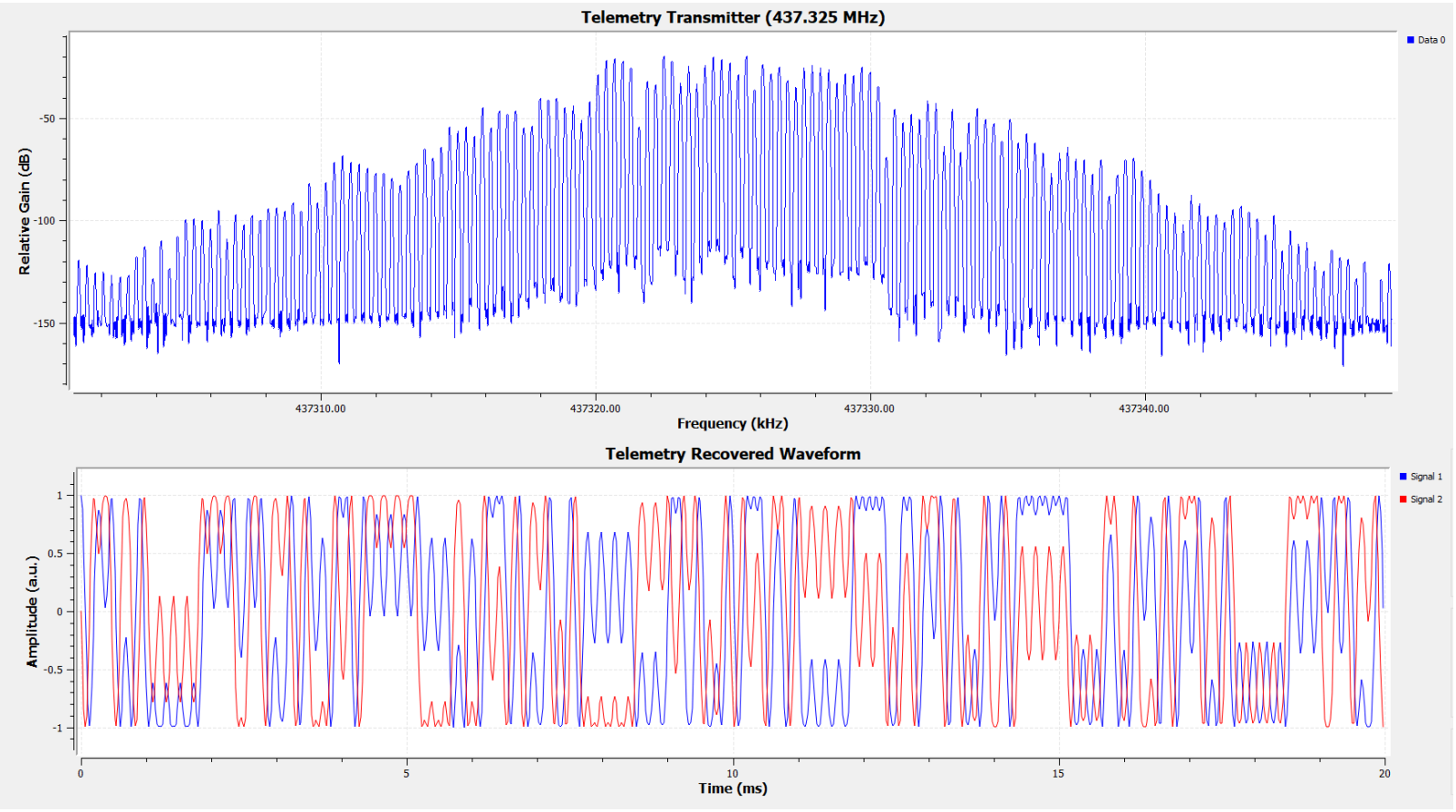


Fig. 1.3 (STL) Telemetry transmitter spectrum and TX waveform (437.325 MHz GFSK)

Ground Station (GND)

The Mode 1 ground station mirrors the satellite link with the transmit and receive roles reversed. Telecommand is the uplink and is generated on the ground as a constant envelope BFSK signal at 437.100 MHz. Telemetry is the downlink and is received and demodulated on the ground at 437.325 MHz. Mode 1 carries no data channel and no beacon, so the applicable figure set is the telecommand transmit and telemetry receive signals.

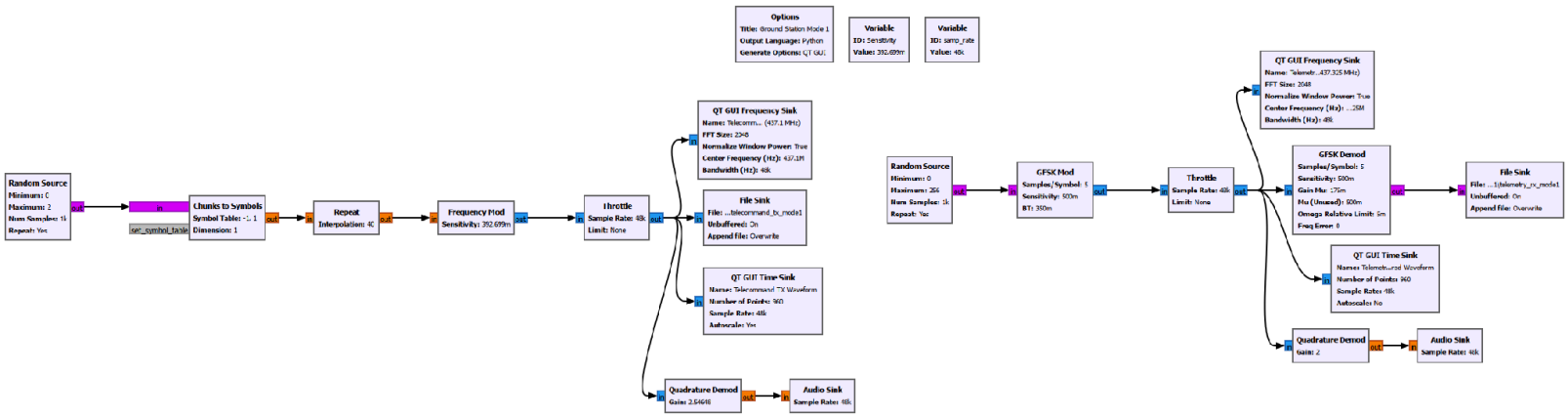


Fig. 1.4 Ground station (GND) flowgraph canvas.

GNU Radio Runtime Plots (GND)

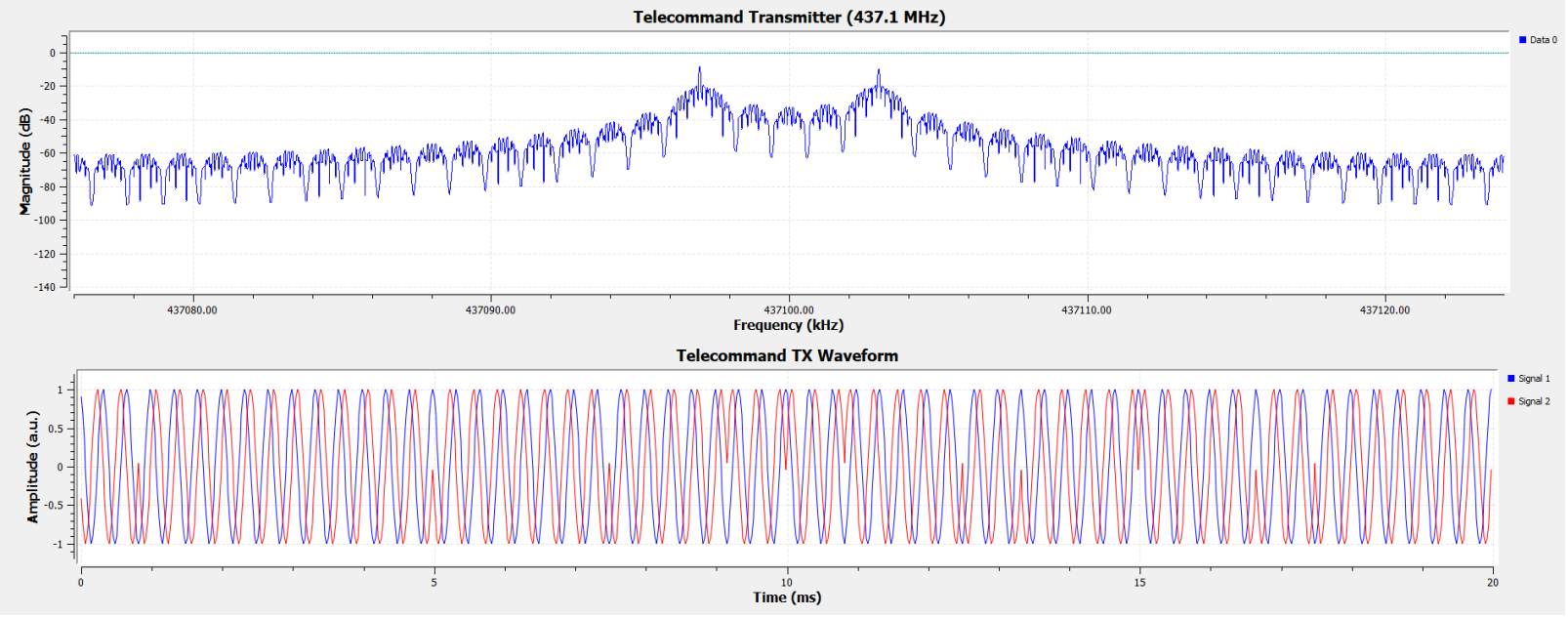


Fig. 1.5 (GND) Telecommand transmitter spectrum and TX waveform (437.1 MHz BFSK)

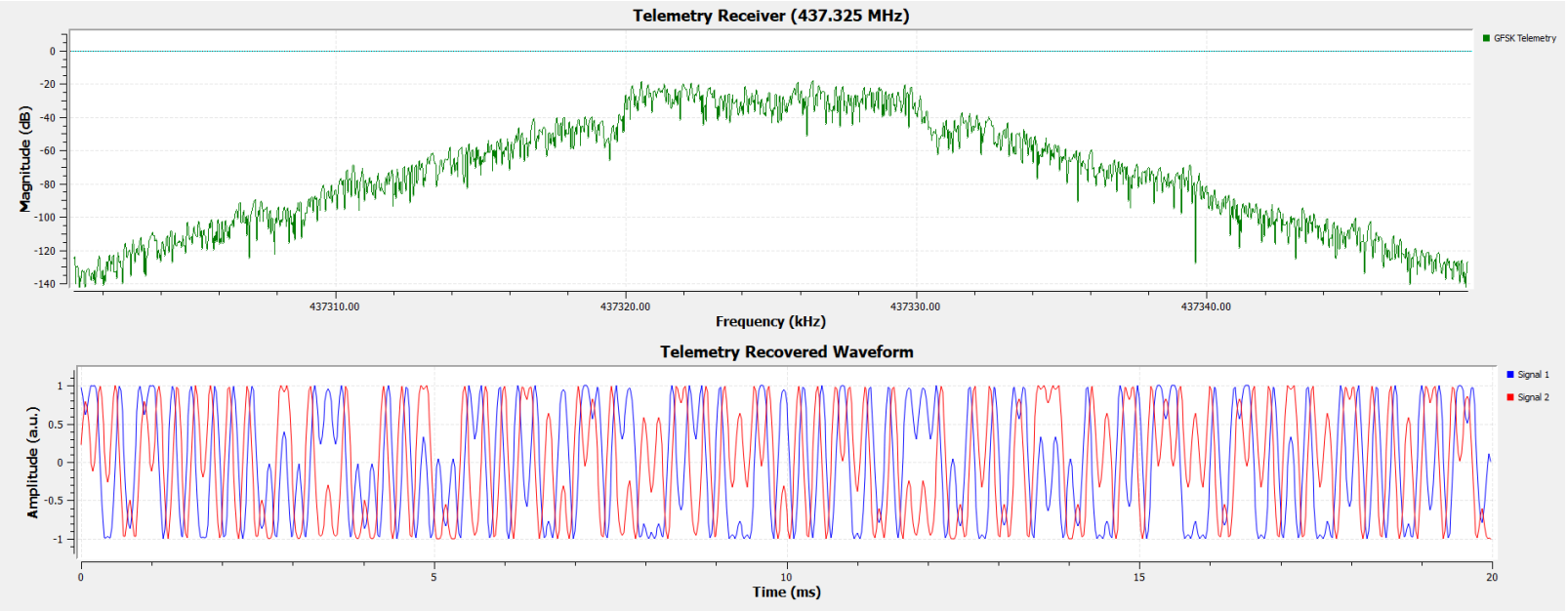


Fig. 1.6 (GND) Telemetry receiver spectrum and recovered waveform (437.325 MHz GFSK)

4.2 Mode 2: General / UHF plus Data Channel

Satellite Terminal (STL)

Mode 2 adds a dual channel UHF payload downlink and a VHF beacon to the Mode 1 baseline. The command and telemetry paths are unchanged, using BFSK at 437.100 MHz and GFSK at 437.325 MHz. The payload data channel uses quadrature phase shift keying in a frequency division duplex arrangement, implemented as two independent constellation modulator chains at 436.5 MHz and 435.8 MHz. Each chain uses differential encoding, 5 samples per symbol, and an excess bandwidth of 0.35, and each is recovered through a channel model, automatic gain control, a polyphase clock synchronizer, and a Costas loop before constellation and differential decoding.

The beacon is a continuous wave carrier at 145.840 MHz keyed at 8.333 Hz, equivalent to 20 words per minute. It is produced by a repeated on-and-off vector, low-pass shaping, and conversion to complex baseband, then detected by a magnitude and threshold chain.

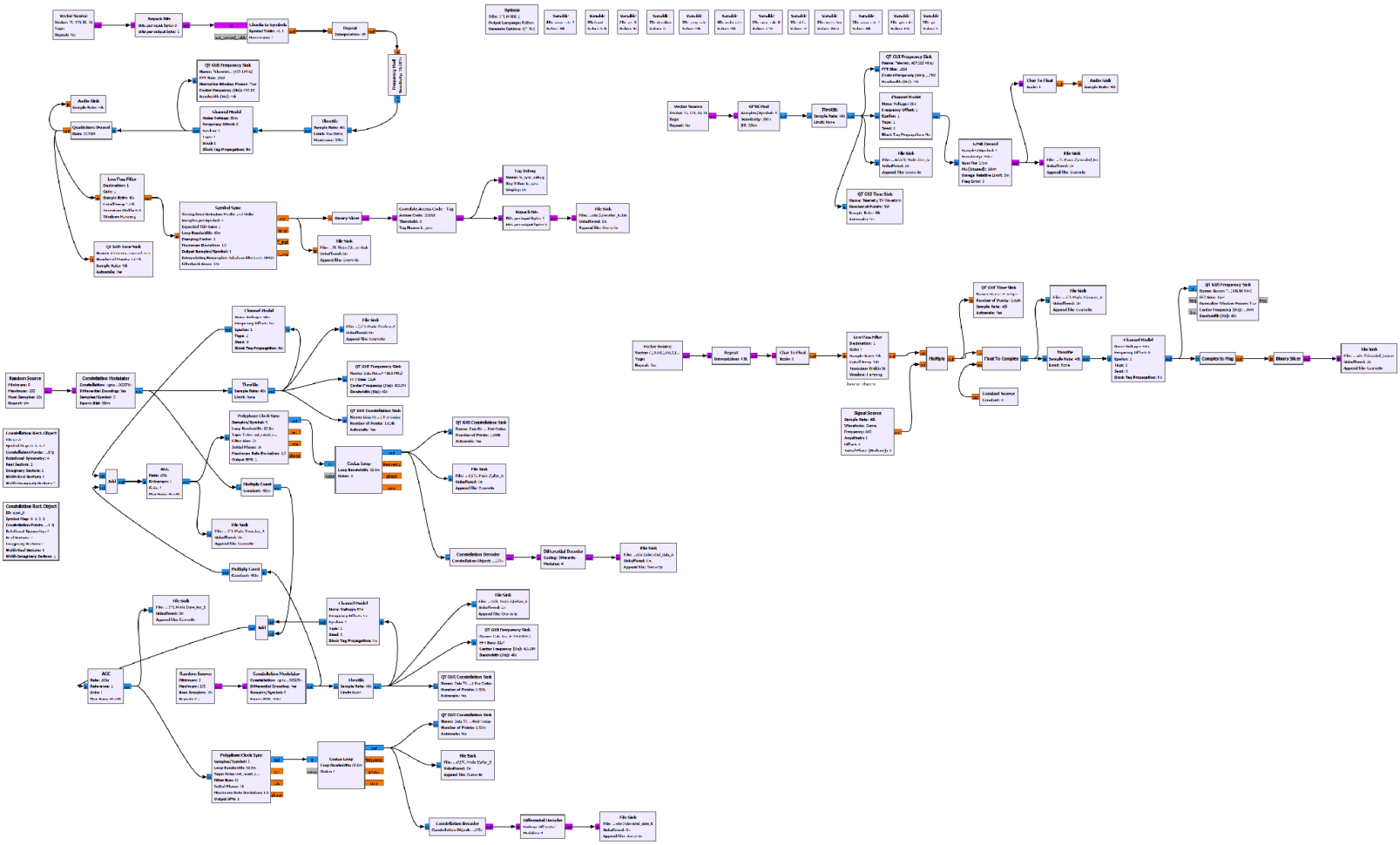


Fig. 2.1 Satellite (STL) flowgraph canvas.

GNU Radio Runtime Plots (STL)

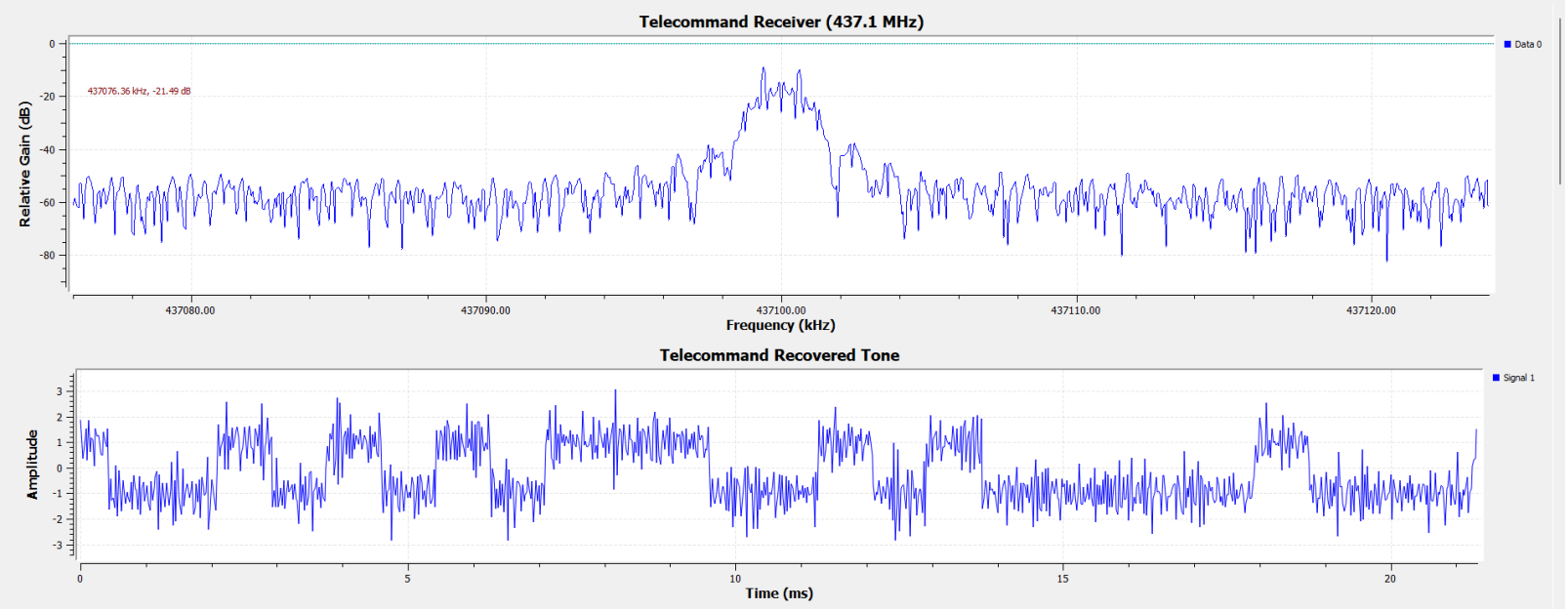


Fig. 2.2 (STL) Telecommand receiver spectrum and recovered tone (437.1 MHz BFSK)

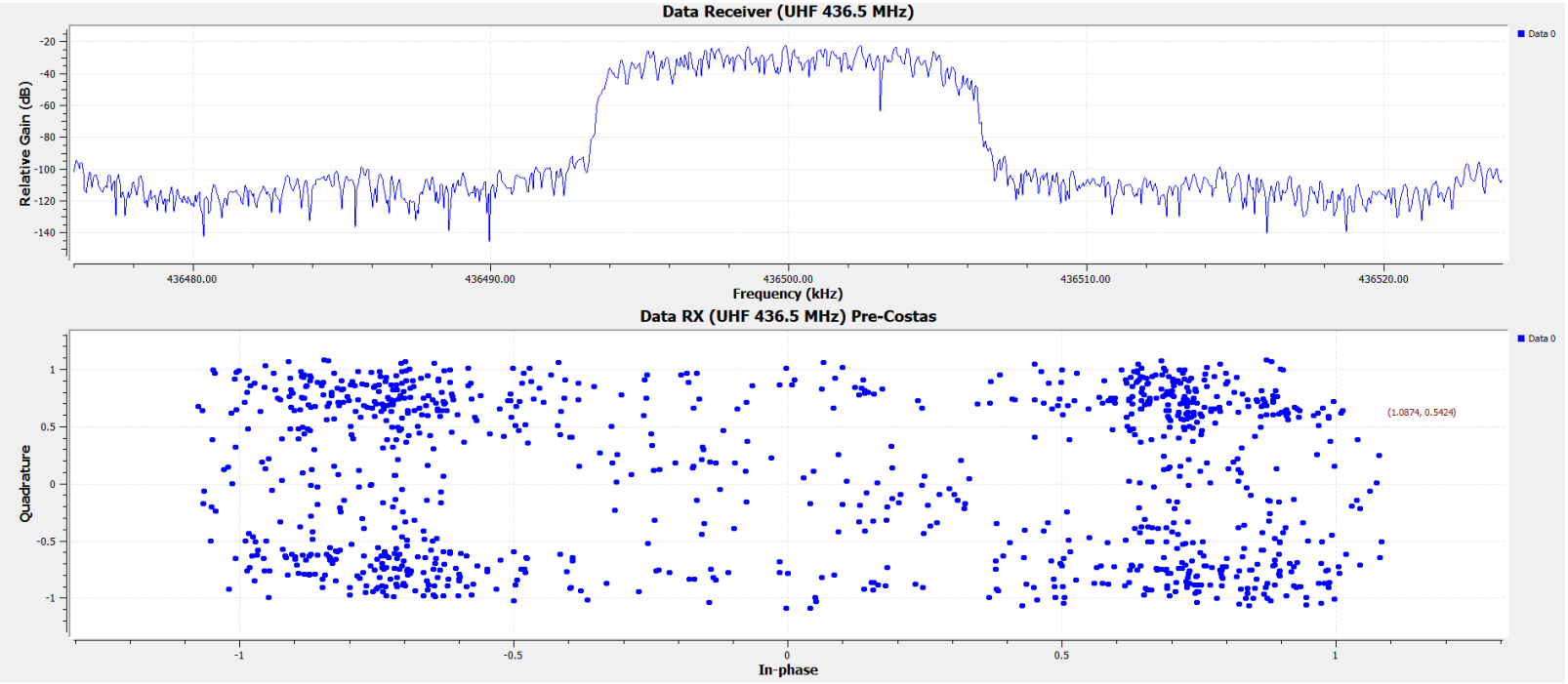


Fig. 2.3 (STL) Telemetry transmitter spectrum and TX waveform (437.325 MHz GFSK)

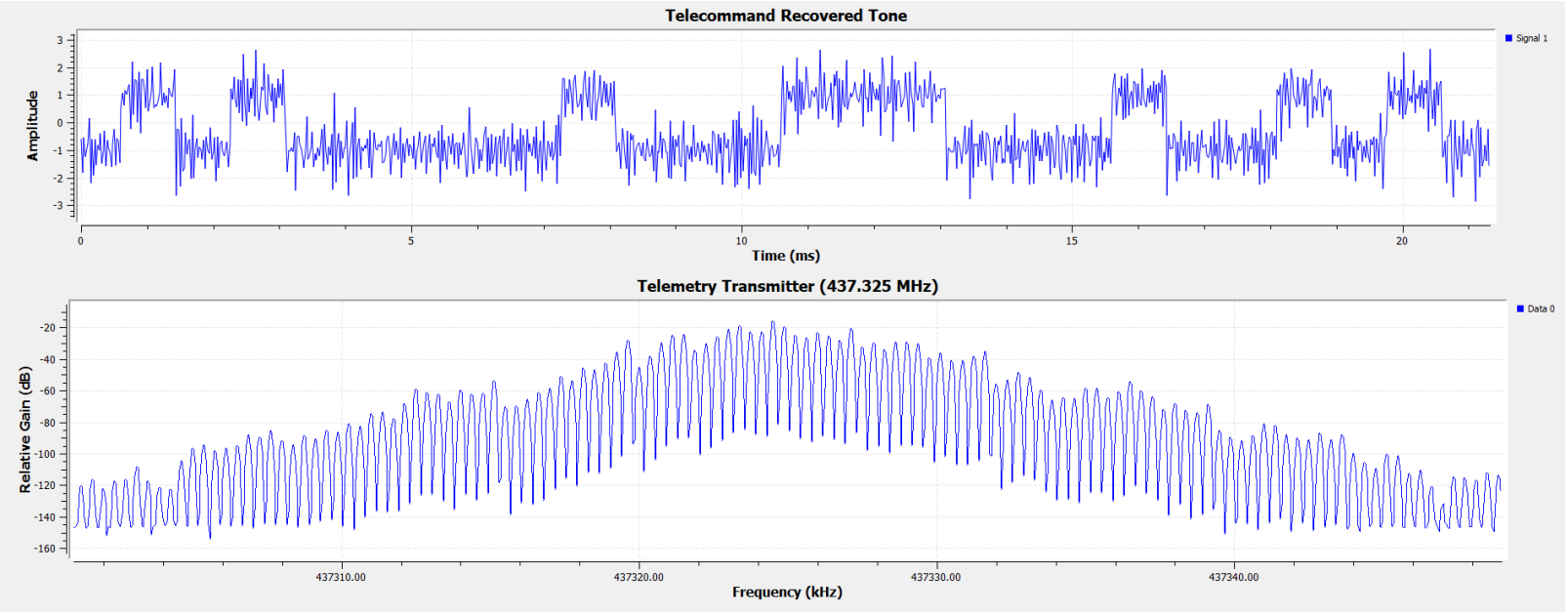


Fig. 2.4 (STL) Data receiver post-Costas constellation and TX spectrum (UHF 436.5 / 435.8 MHz)

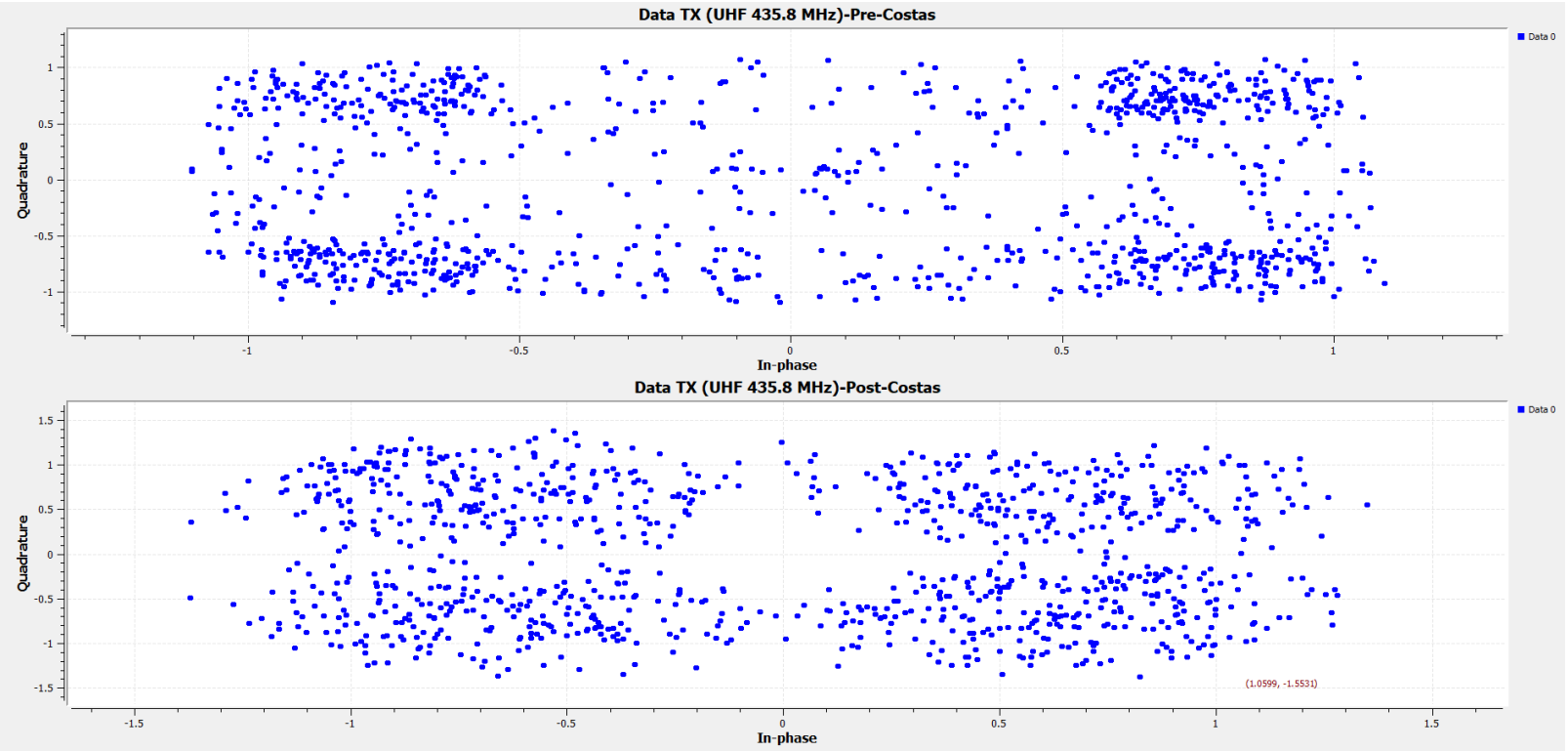


Fig. 2.5 (STL) Beacon transmitter spectrum and envelope (145.84 MHz CW)

Ground Station (GND)

The Mode 2 ground station receives both payload data subchannels and the beacon in addition to the command and telemetry links. The QPSK data receiver recovers a clean four-point constellation with balanced dibit statistics. The ground beacon receiver recovers the 8.333 Hz keying as a clean on and off envelope, which confirms that the beacon downlink closes on the ground segment. Telecommand is transmitted from the ground as constant envelope BFSK, and telemetry is received at 437.325 MHz.

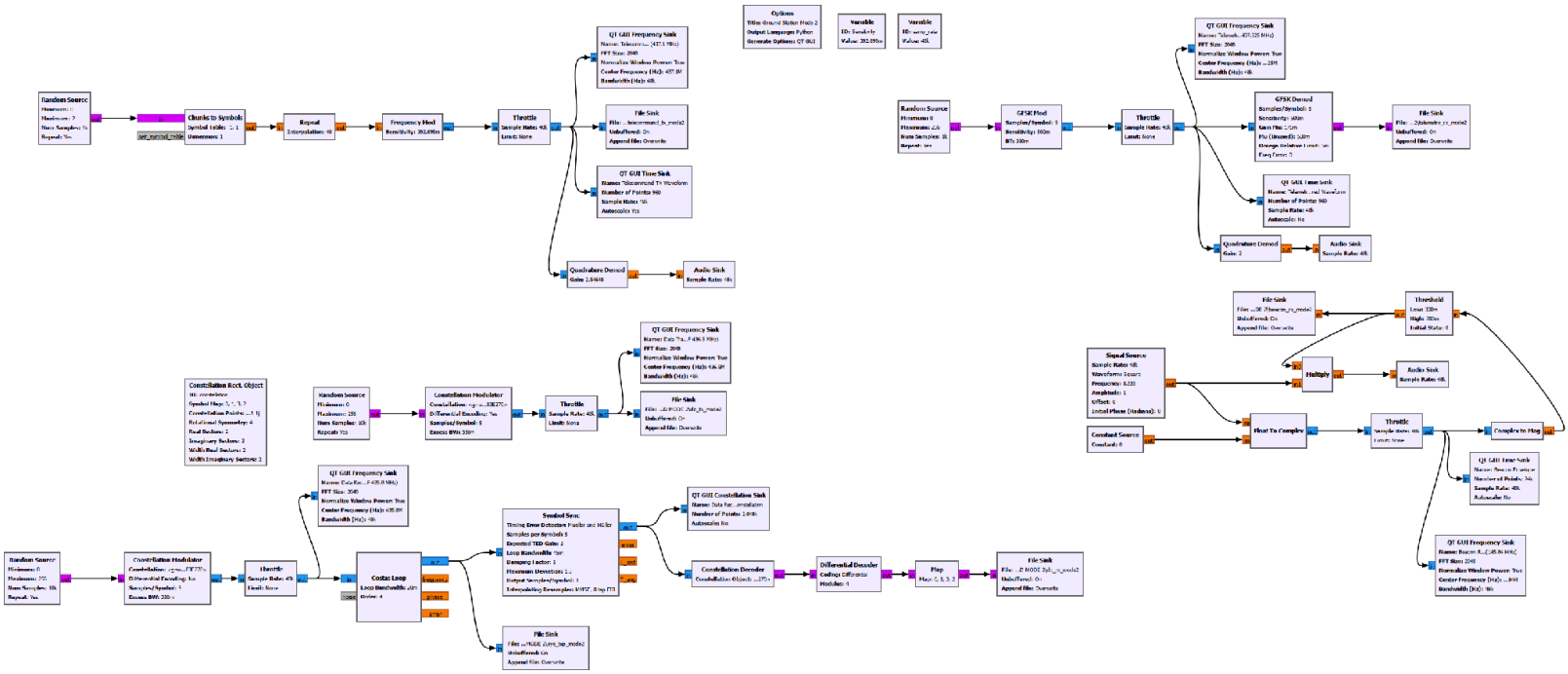


Fig. 2.6 Ground station (GND) flowgraph canvas.

GNU Radio Runtime Plots (GND)

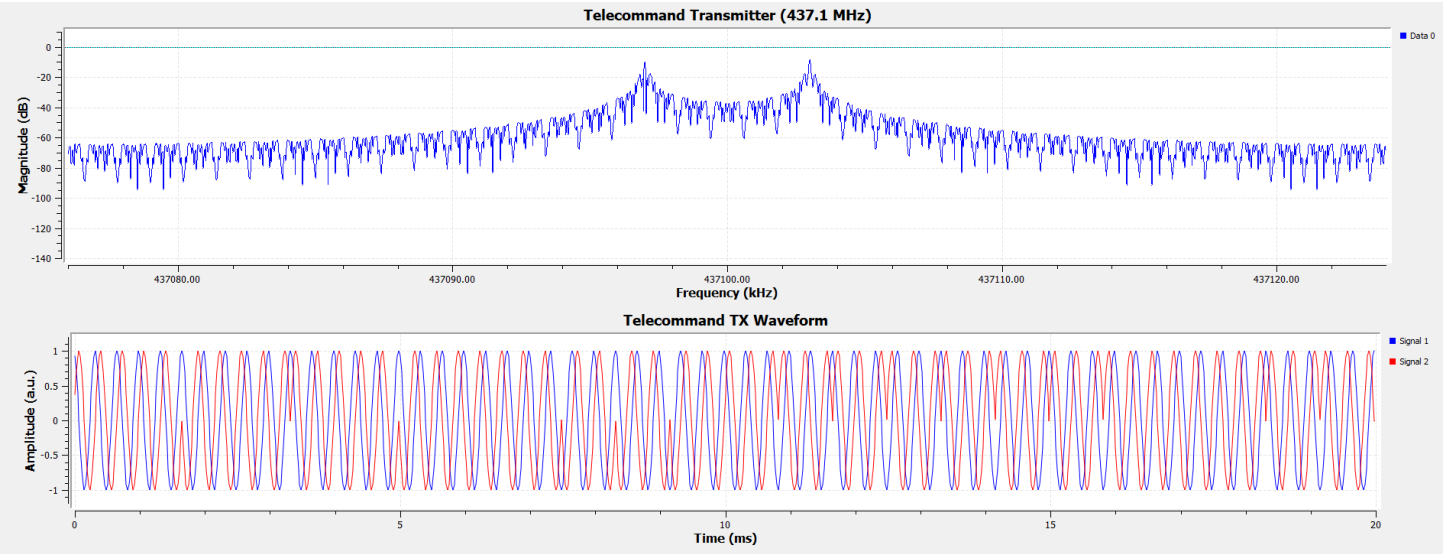


Fig. 2.7 (GND) Telecommand transmitter spectrum and TX waveform (437.1 MHz BFSK)

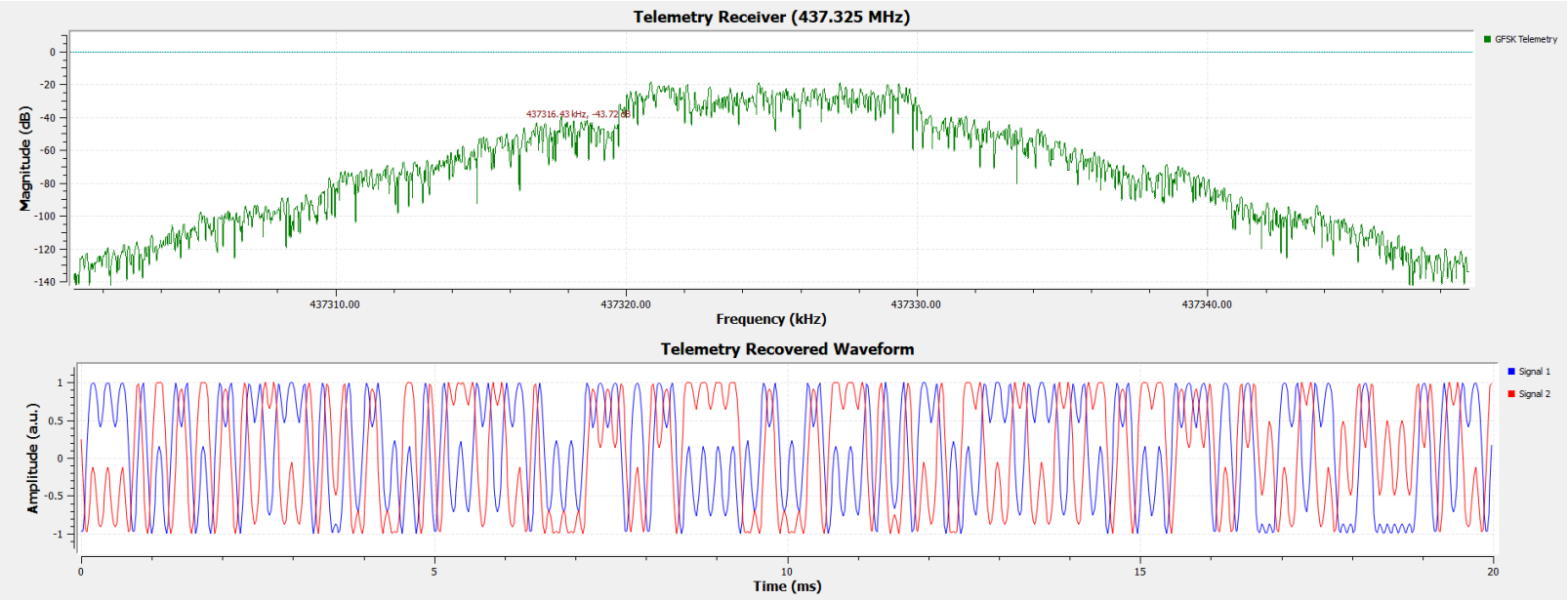


Fig. 2.8 (GND) Telemetry receiver spectrum and recovered waveform (437.325 MHz GFSK)

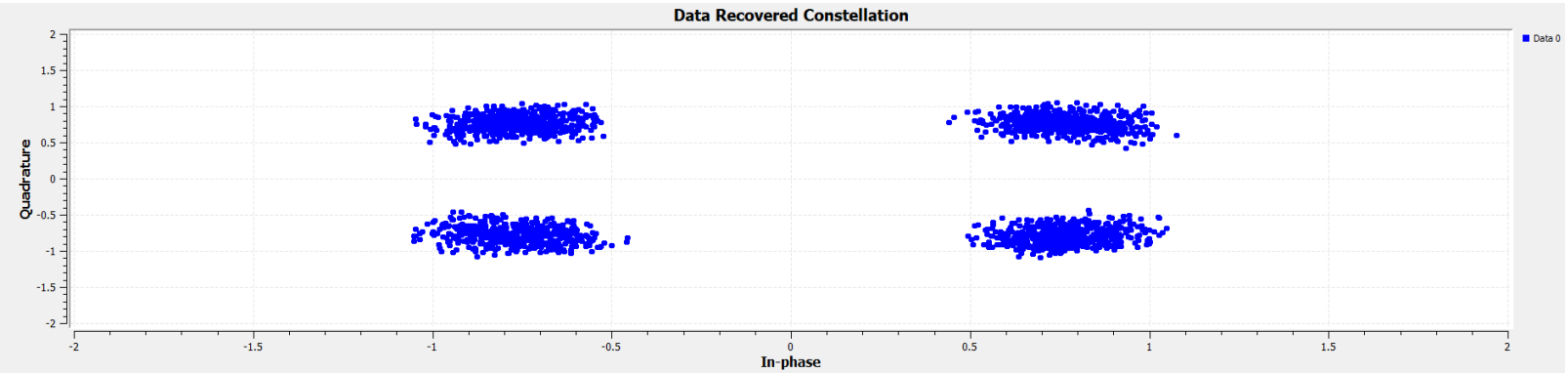


Fig. 2.9 (GND) Data-recovered constellation, four QPSK lobes (UHF data channel)

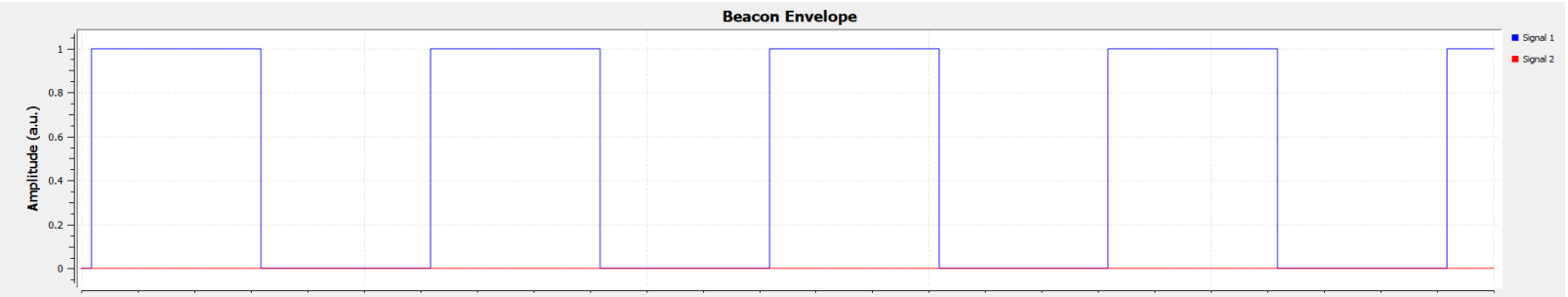


Fig. 2.10 (GND) Beacon envelope, recovered on-off keying (145.84 MHz)

4.3 Mode 3: Inter-Satellite Link (ISL)

Satellite Terminal (STL)

Mode 3 provides the inter-satellite link configuration. The command, telemetry, and beacon paths are unchanged from Mode 2, but the payload data channel is replaced by a single S-band QPSK link operating in time division duplex at 2245 MHz. The S-band chain runs at a sample rate of 2 MHz with 500 kilosymbols per second, 4 samples per symbol, and an excess bandwidth of 0.35.

The constellation uses the standard mapping with points at the four diagonal positions and a symbol map of 0, 1, 3, and 2 and a rotational symmetry of 4. Recovery is performed through a polyphase clock synchronizer, a Costas loop, a constellation decoder, and a differential decoder. The runtime plots include the TDD burst envelope, which shows the time-division duplex on and off bursts that define this mode.

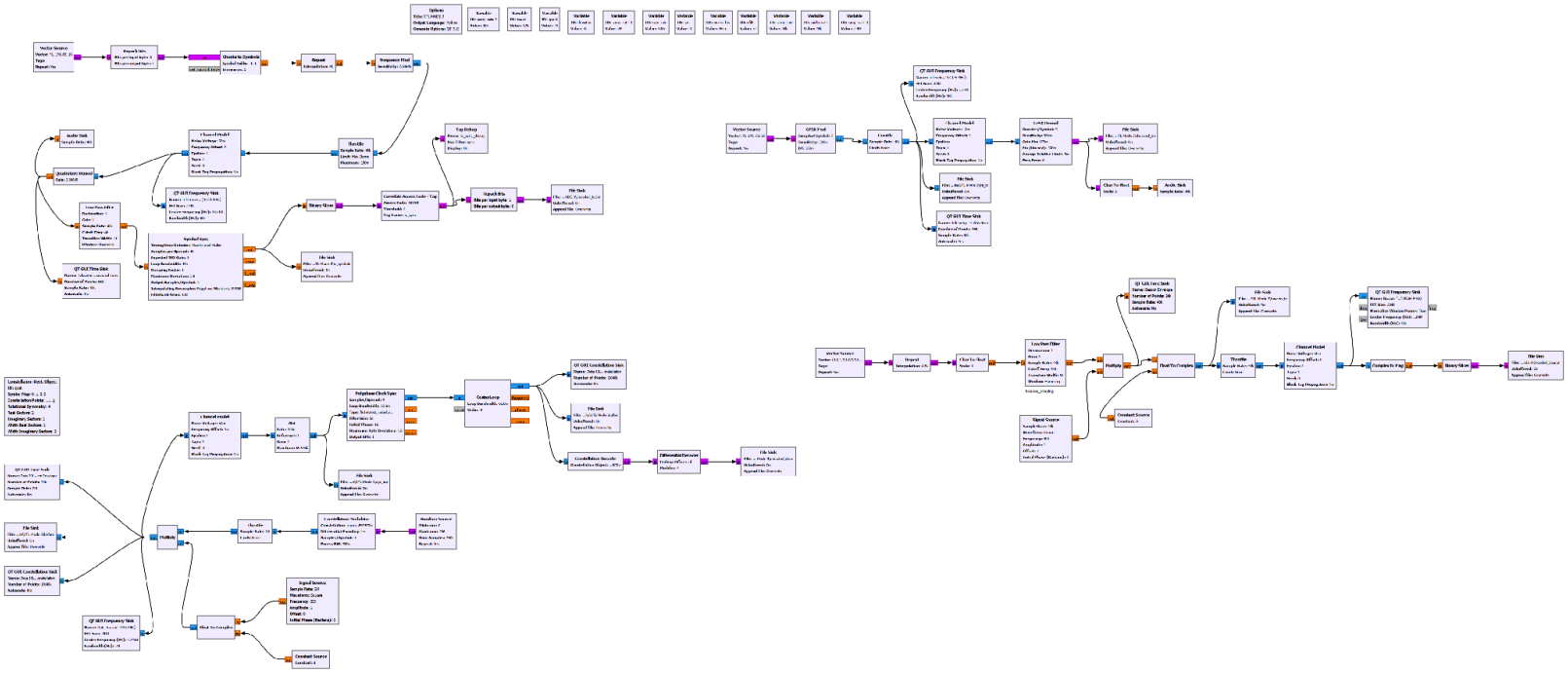


Fig. 3.1 Satellite (STL) flowgraph canvas.

GNU Radio Runtime Plots (STL)

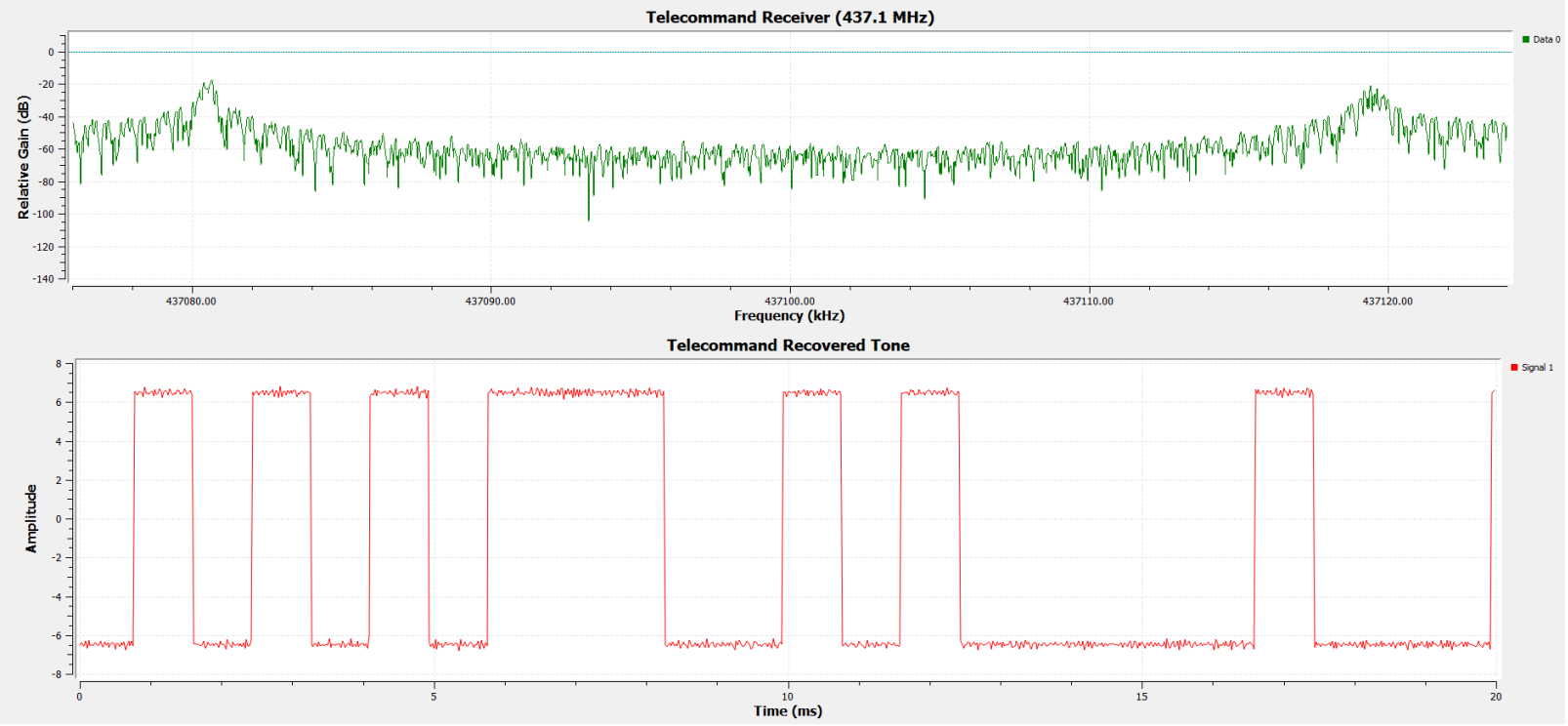


Fig. 3.2 (STL) Telecommand receiver spectrum and recovered tone (437.1 MHz BFSK)

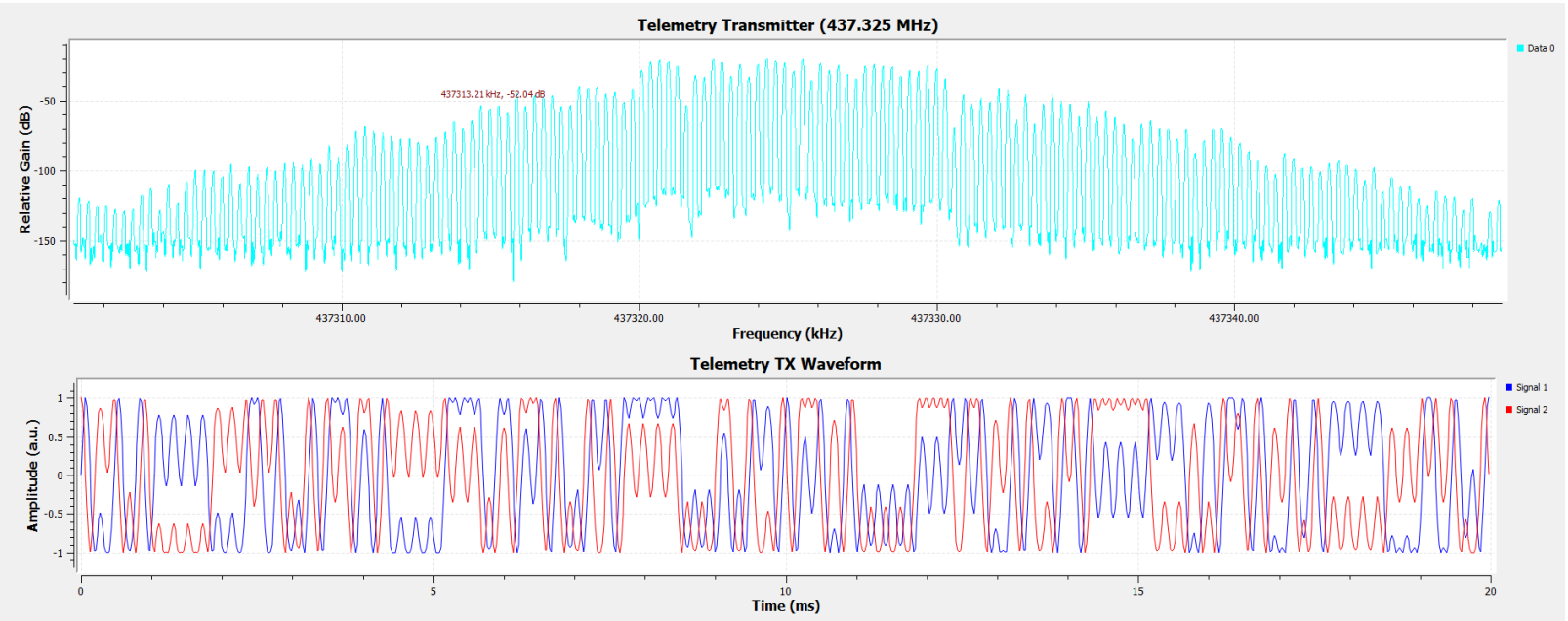


Fig. 3.3 (STL) Telemetry transmitter spectrum and TX waveform (437.325 MHz GFSK)

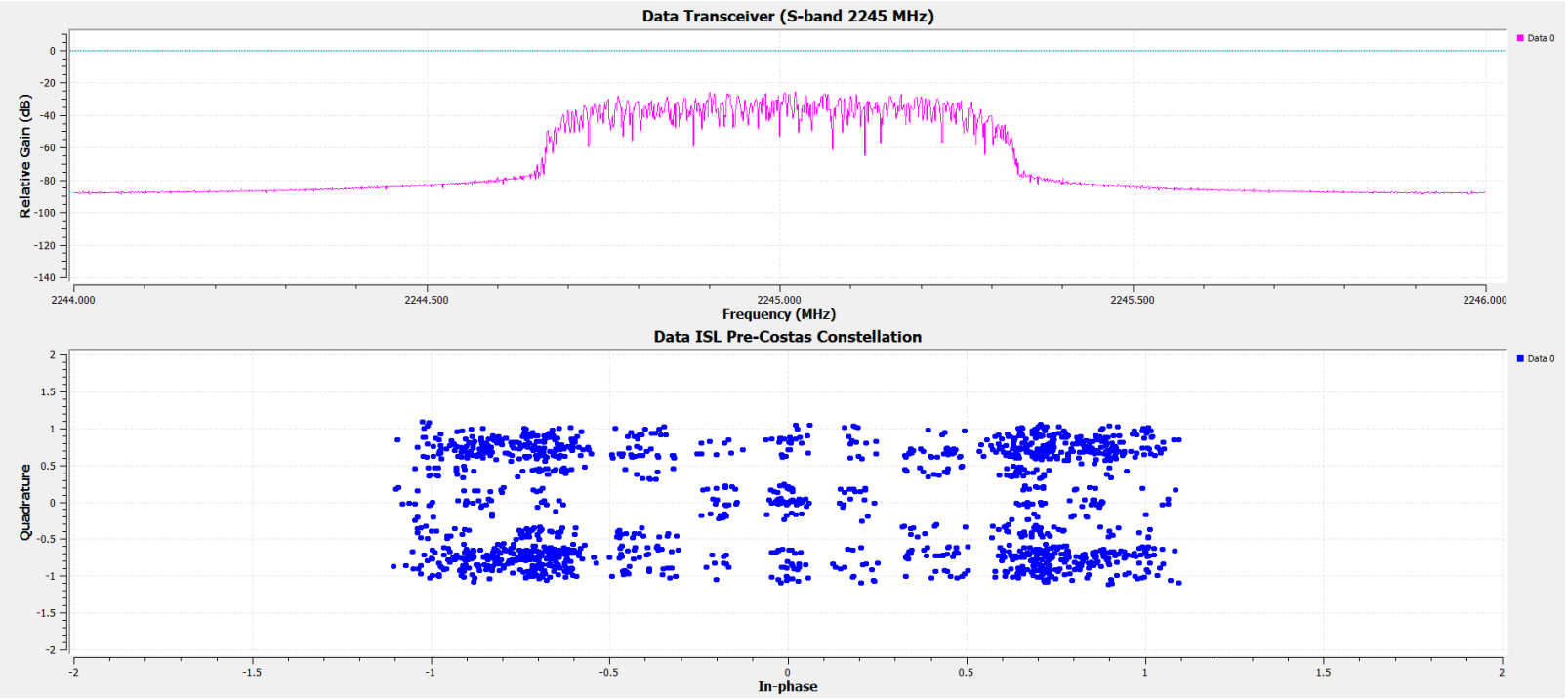


Fig. 3.4 (STL) Data transceiver S-band spectrum and pre-Costas constellation (2245 MHz)

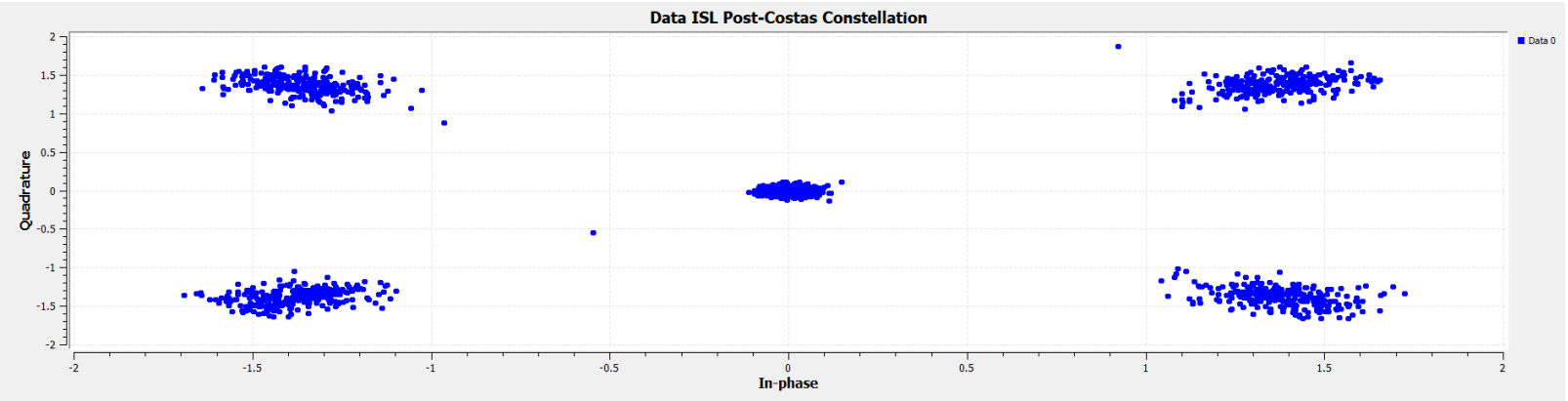


Fig. 3.5 (STL) Data ISL post-Costas constellation, four recovered QPSK lobes (2245 MHz)

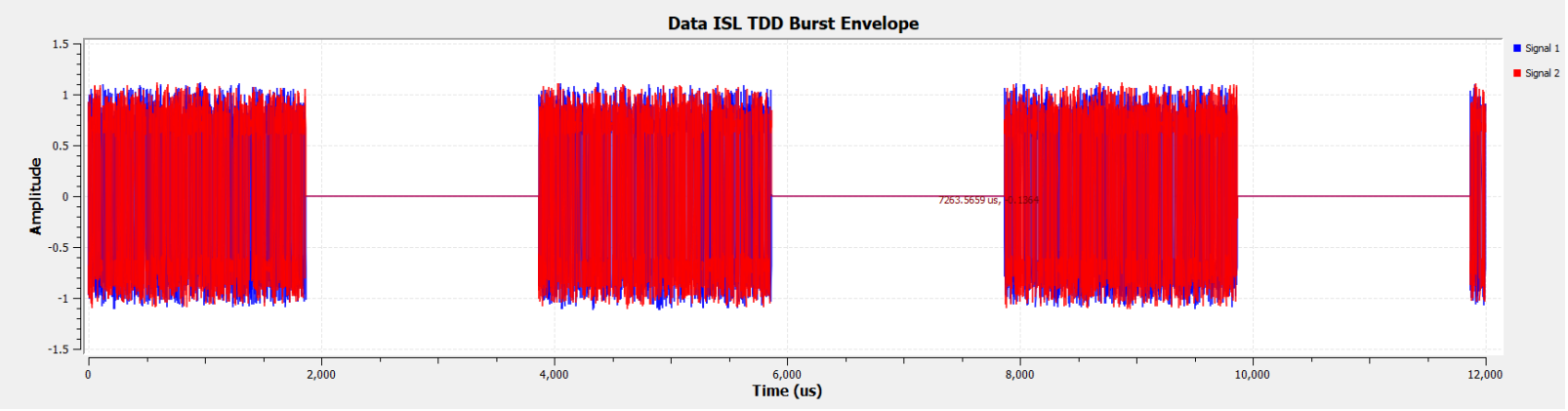


Fig. 3.6 (STL) Data ISL TDD burst envelope, showing time-division duplex operation

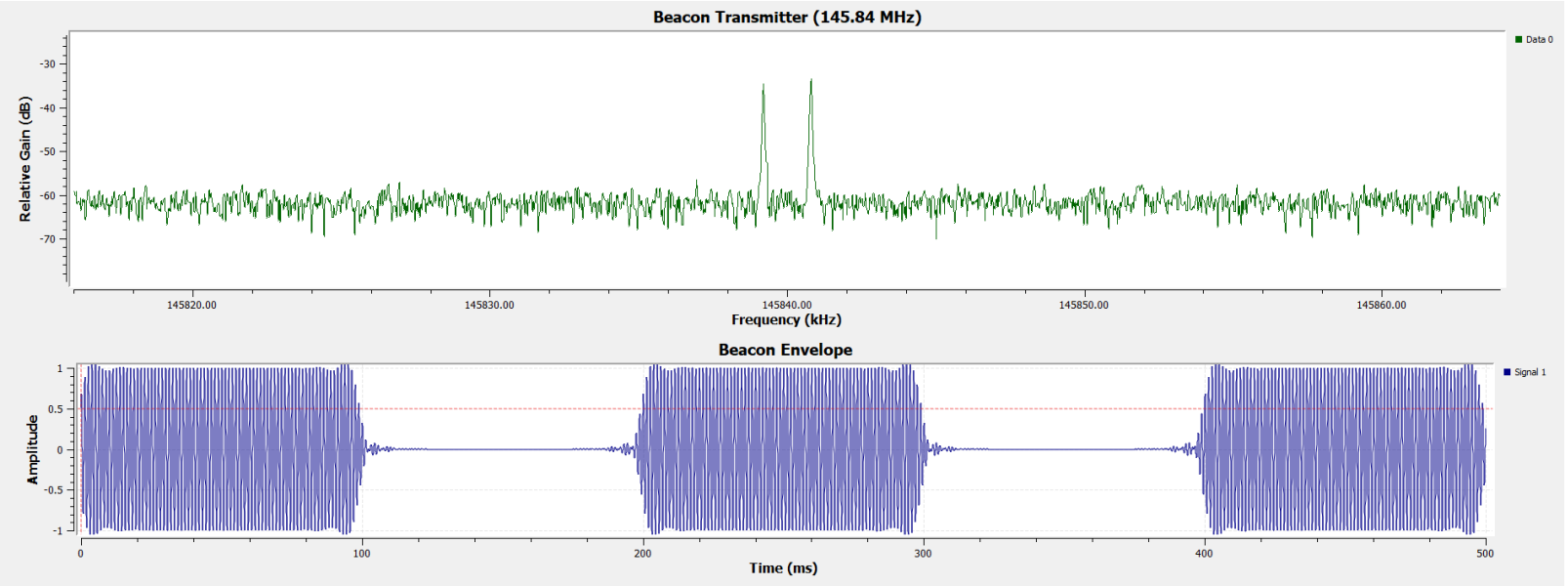


Fig. 3.7 (STL) Beacon transmitter spectrum and envelope (145.84 MHz CW)

Ground Station (GND)

The Mode 3 ground station captures the command uplink, the telemetry downlink, and the VHF beacon. The recovered telemetry is balanced, and the telecommand transmit signal shows the expected constant envelope BFSK structure. The beacon receiver recovers the 8.333 Hz keying cleanly. The S-band inter-satellite data channel is not part of this ground capture, which is consistent with the ground station table where the Mode 3 data column is zero, so the ground figure set covers the command, telemetry, and beacon signals in two-dimensional and three-dimensional views.

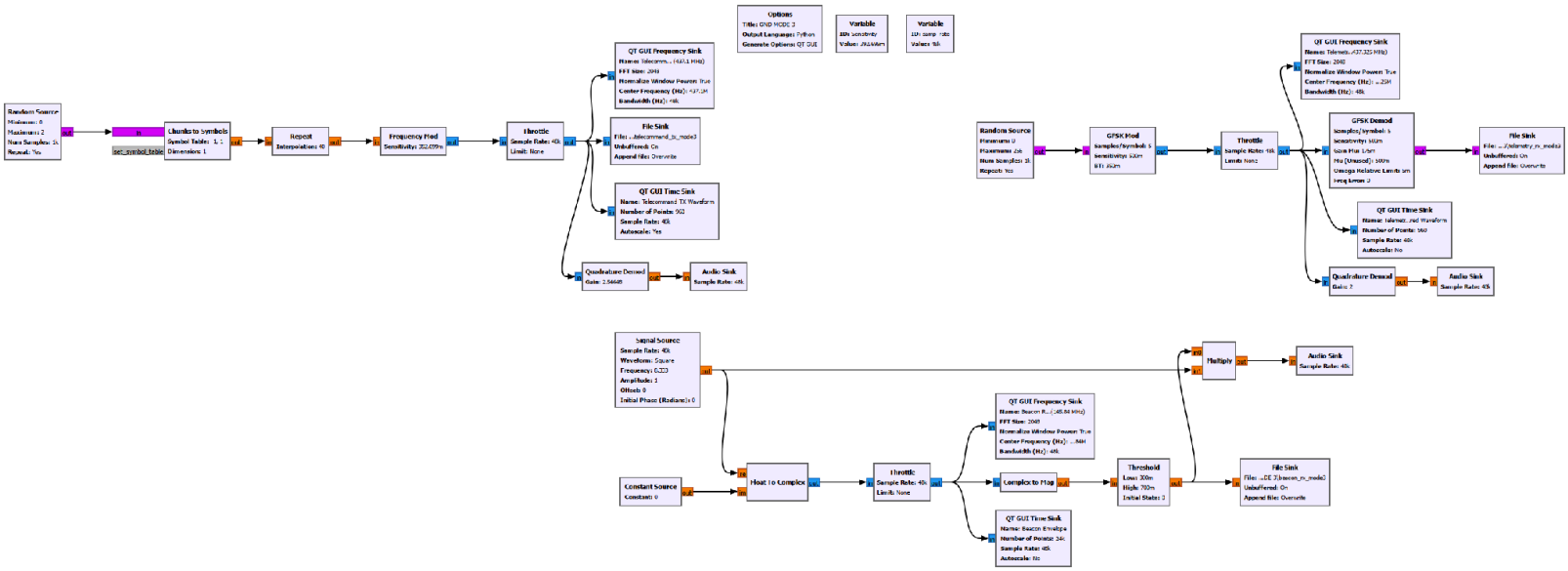


Fig. 3.8 Ground station (GND) flowgraph canvas.

Validation note. All three ground file sinks route correctly to the Mode 3 output directory. The command, telemetry, and beacon links are healthy, and no routing errors were found in this canvas.

GNU Radio Runtime Plots (GND)

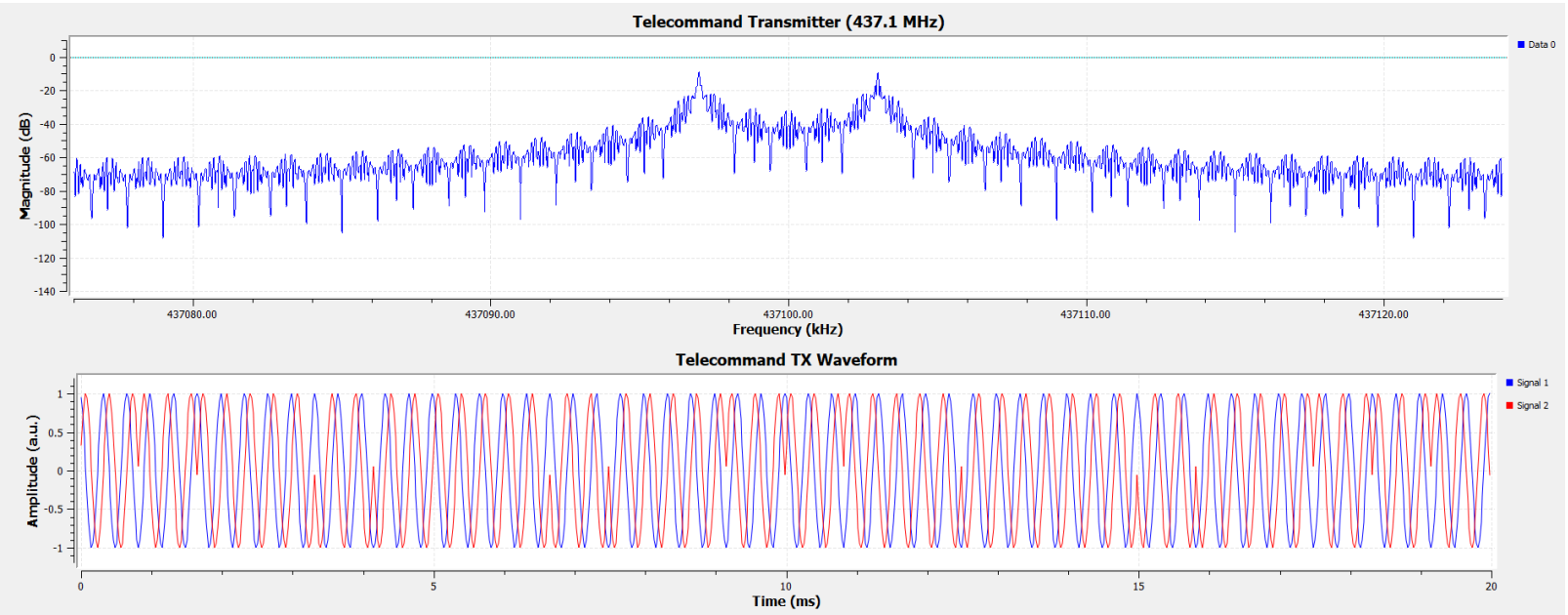


Fig. 3.9 (GND) Telecommand transmitter spectrum and TX waveform (437.1 MHz BFSK)



Fig. 3.10 (GND) Telemetry receiver spectrum and recovered waveform (437.325 MHz GFSK)

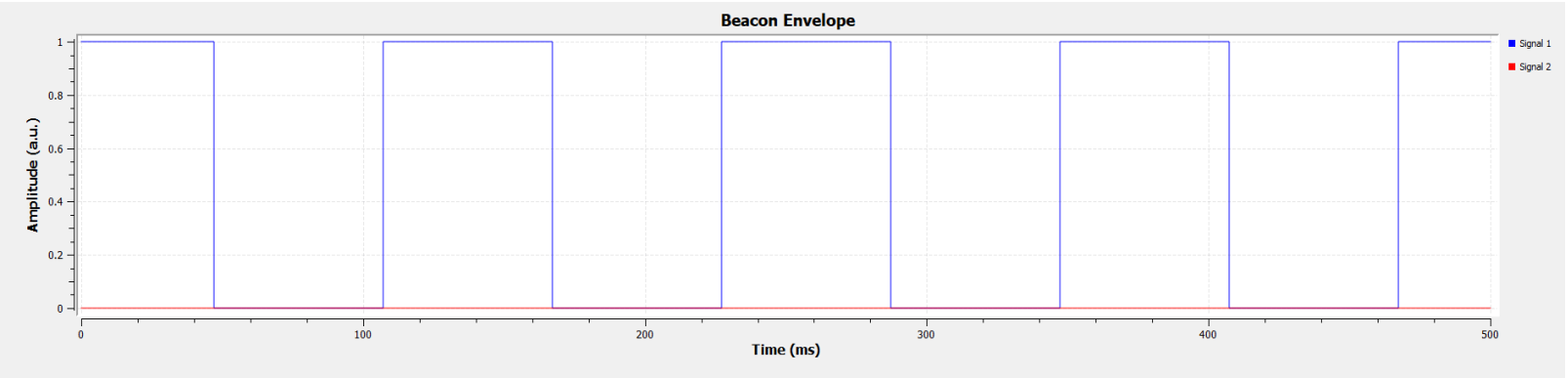


Fig. 3.11 (GND) Beacon envelope, recovered on-off keying (145.84 MHz)

Recovered Signal Figures (offline analysis)

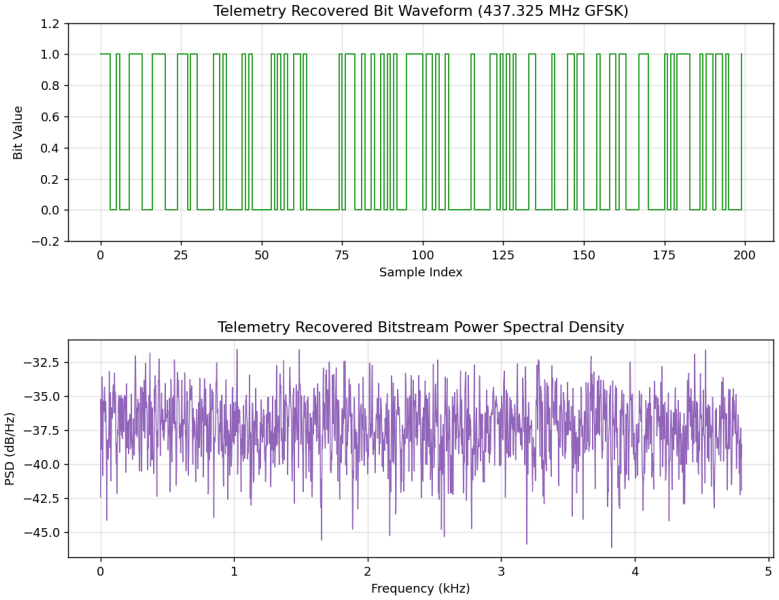


Fig. 3.12 Telemetry RX bit waveform and PSD (437.325 MHz GFSK)

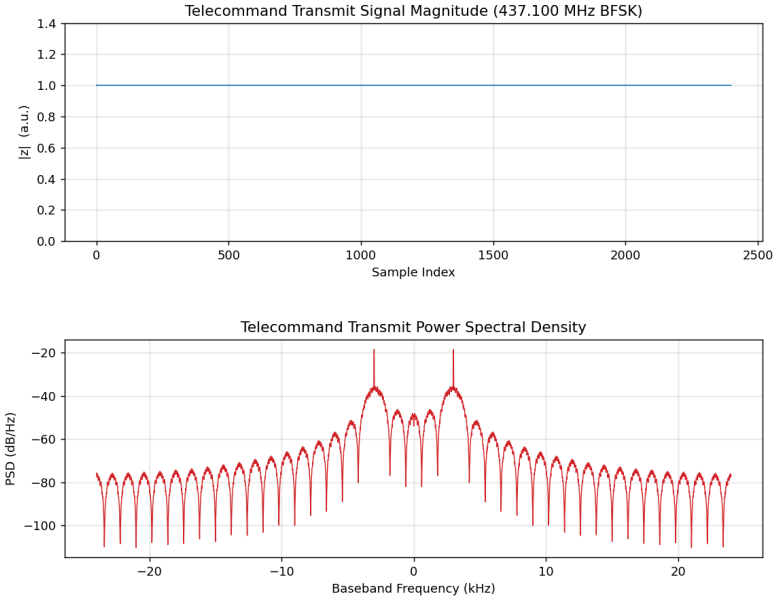


Fig. 3.13: Telecommand TX magnitude and PSD (437.100 MHz BFSK)

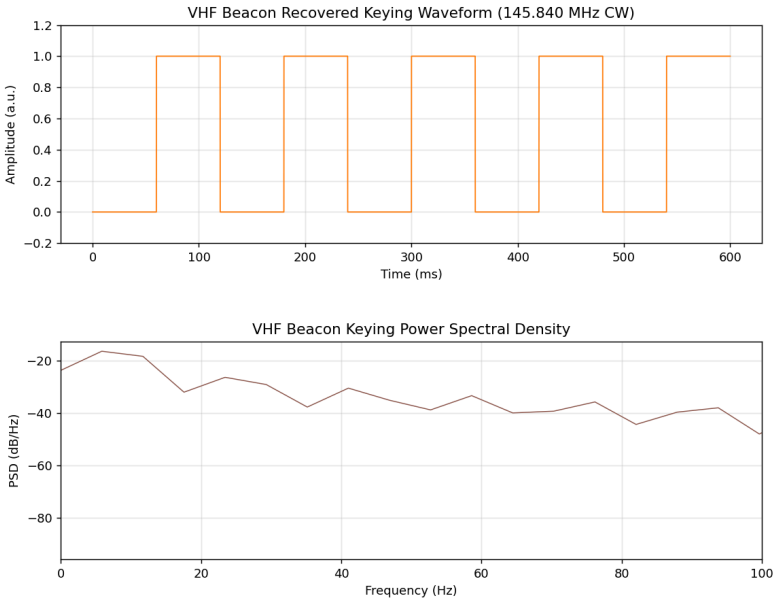


Fig. 3.14 VHF beacon keying waveform and PSD (145.840 MHz CW)

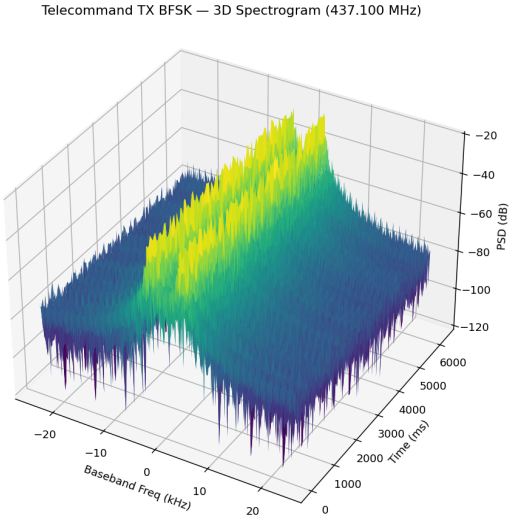


Fig. 3.15 Telecommand BFSK 3D spectrogram

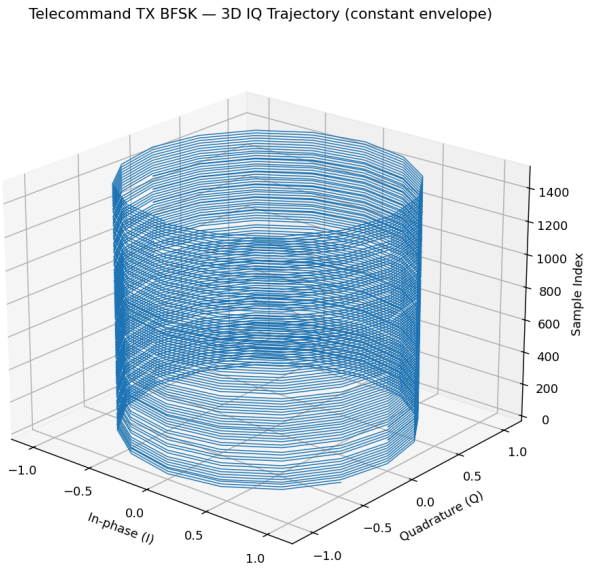


Fig. 3.16 Telecommand BFSK 3D IQ trajectory

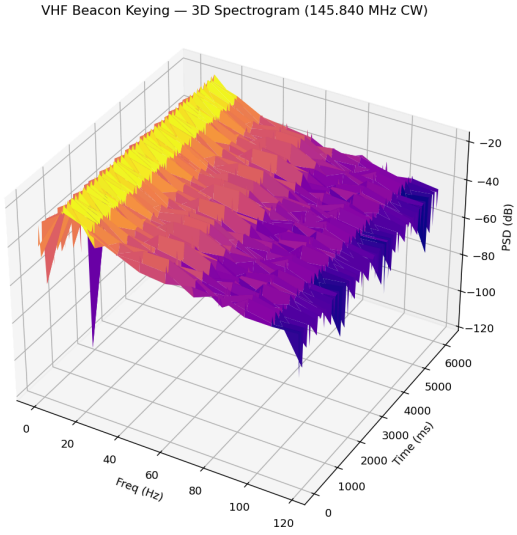


Fig. 3.17 VHF beacon keying 3D spectrogram

Four-View Telecommand Spectrogram

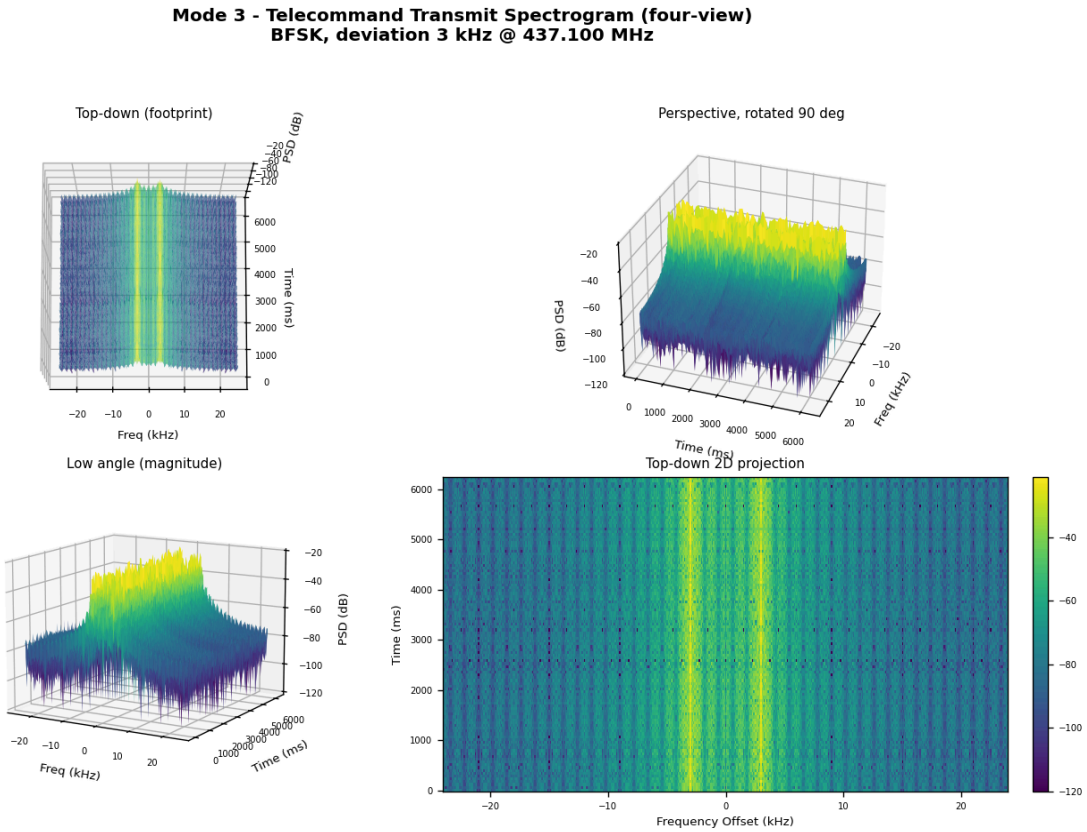


Fig. 3.18. Telecommand transmit spectrogram shown from four perspectives (top-down, rotated, low-angle, and 2D projection).

4.4 Mode 4: High Data Downlink / S-band

Satellite Terminal (STL)

Mode 4 is the high data downlink configuration. It retains the command, telemetry, and beacon links of Mode 3 and uses a transmit-only S-band QPSK downlink at 2245 MHz for high-rate payload data. The S-band transmit chain uses the same 2 MHz sample rate, 4 samples per symbol, and 0.35 excess bandwidth as Mode 3 but omits the return direction, reflecting a downlink-only high-throughput scenario.

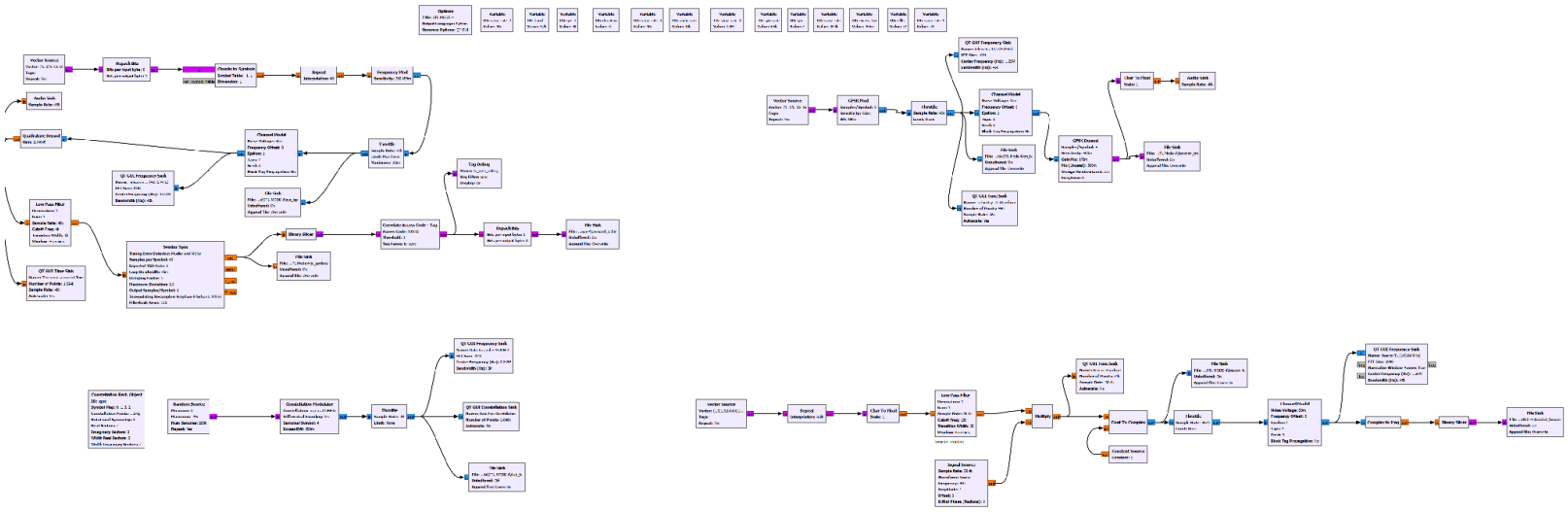


Fig. 4.1 Satellite (STL) flowgraph canvas.

Validation note. The satellite telemetry frequency sink is titled Telemetry Transmitter (VHF 437.325 MHz). The 437.325 MHz carrier is in the UHF band, not VHF, so the VHF label is a copy-paste artifact in the sink name. The signal itself is correct; only the on-screen title is affected.

GNU Radio Runtime Plots (STL)

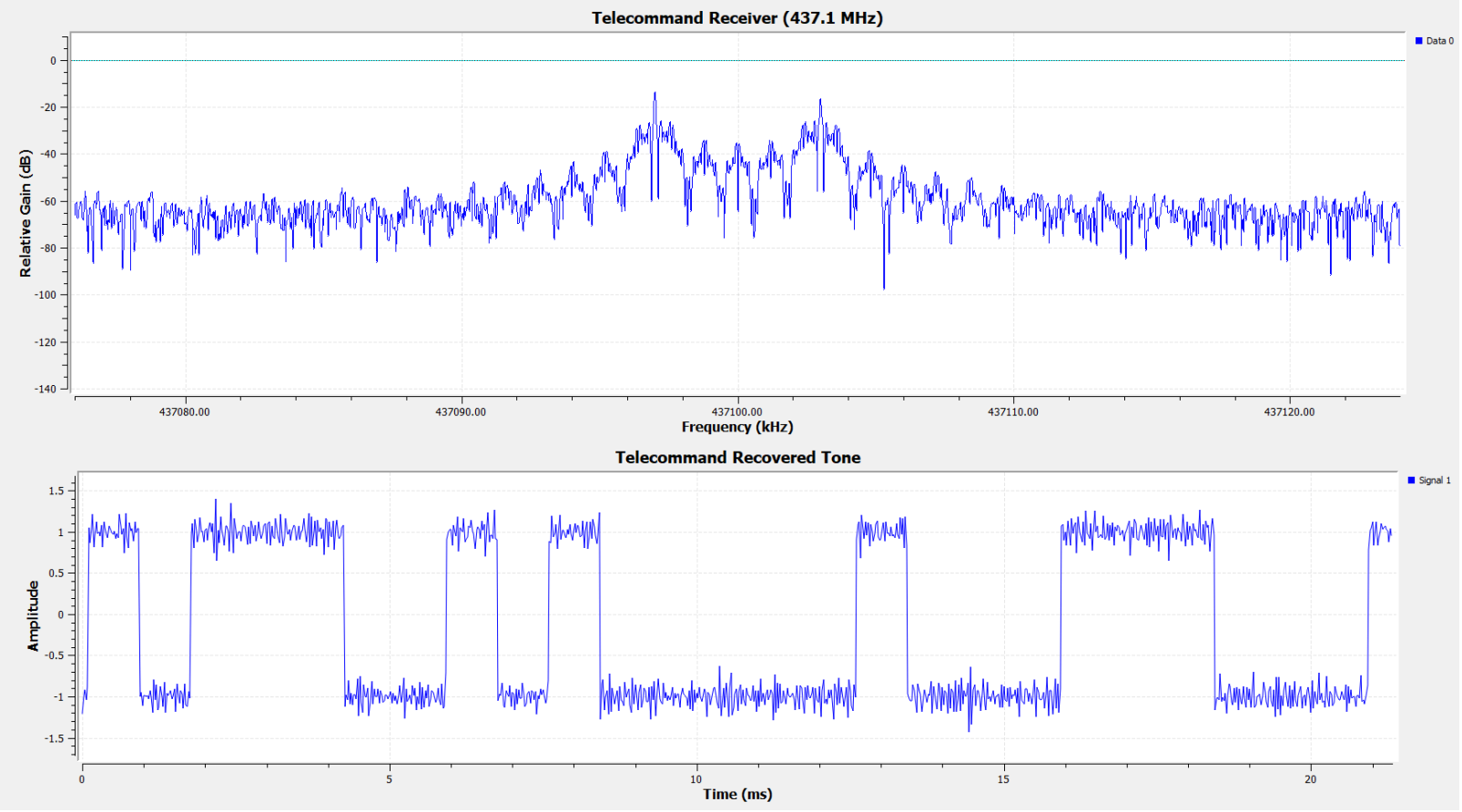


Fig. 4.2 (STL) Telecommand receiver spectrum and recovered tone (437.1 MHz BFSK)

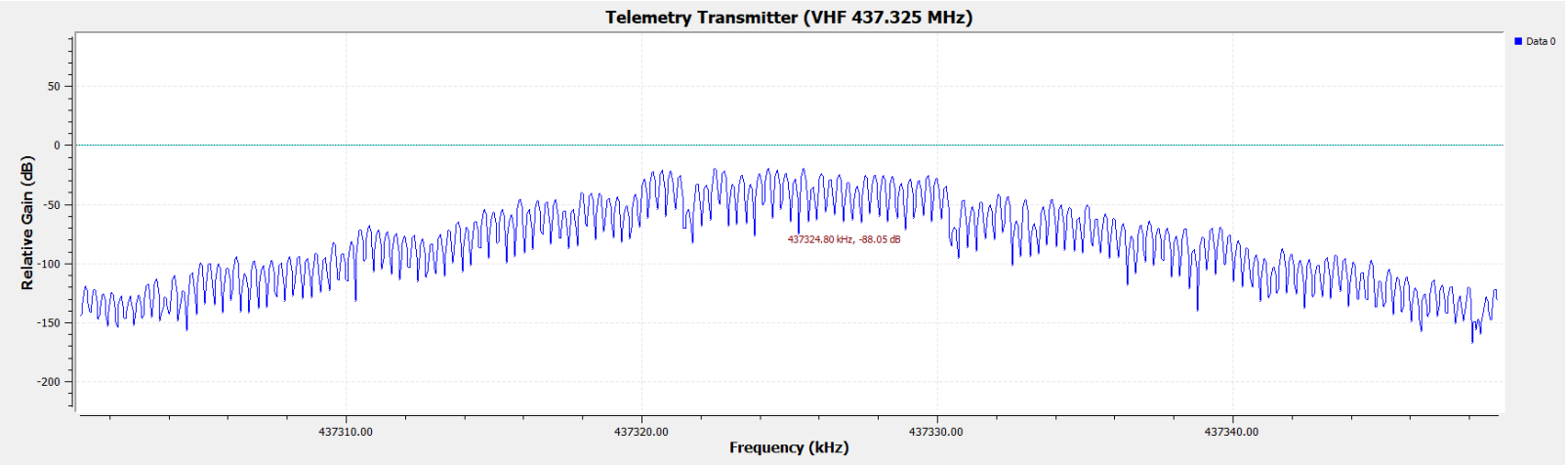


Fig. 4.3 (STL) Telemetry transmitter spectrum (437.325 MHz GFSK; sink title mislabels this as VHF, see note)

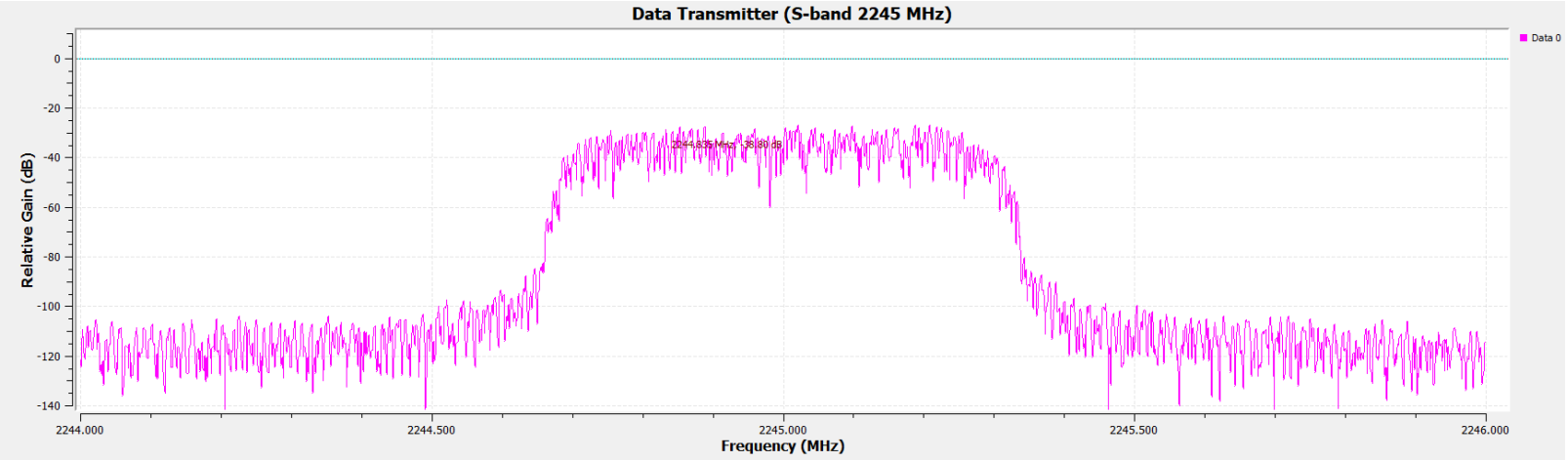


Fig. 4.4 (STL) Data transmitter S-band spectrum, transmit-only downlink (2245 MHz)

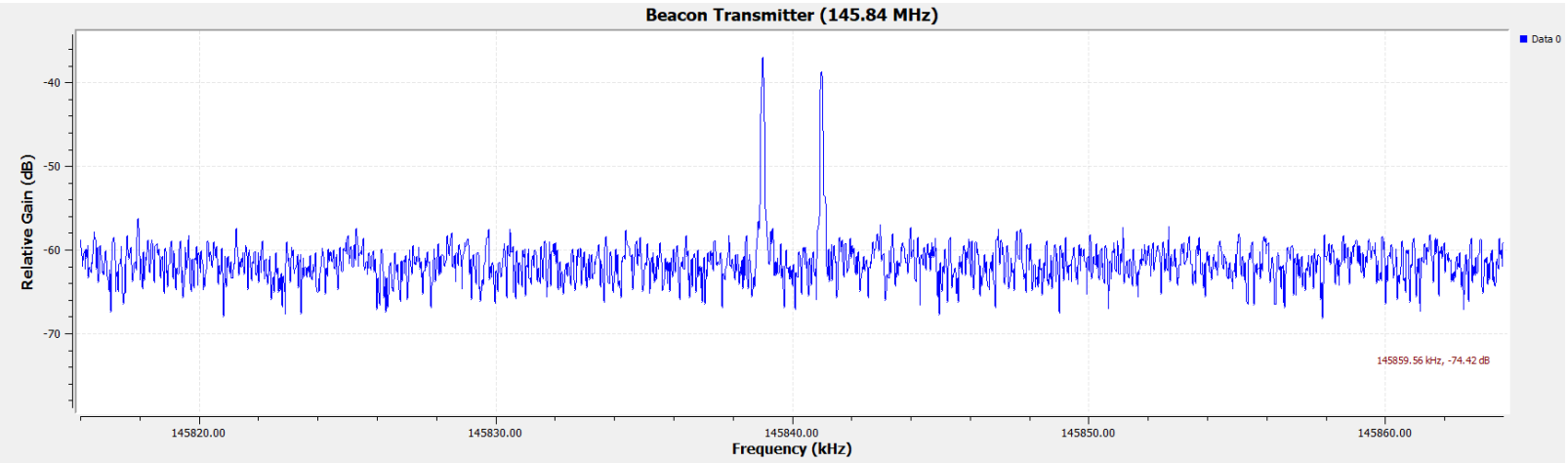


Fig. 4.5 (STL) Beacon transmitter spectrum (145.84 MHz CW)

Ground Station (GND)

The Mode 4 ground station receives the command uplink, the telemetry downlink, the VHF beacon, and the S-band data downlink. The S-band sample rate variable is correctly set to 2 MHz. The recovered data symbols are balanced across all four QPSK states at approximately 25 percent each, which confirms that the S-band decode chain functions correctly. The live GNU Radio runtime constellation recovers four clean QPSK lobes on the ground, which provides the measured constellation evidence for this mode.

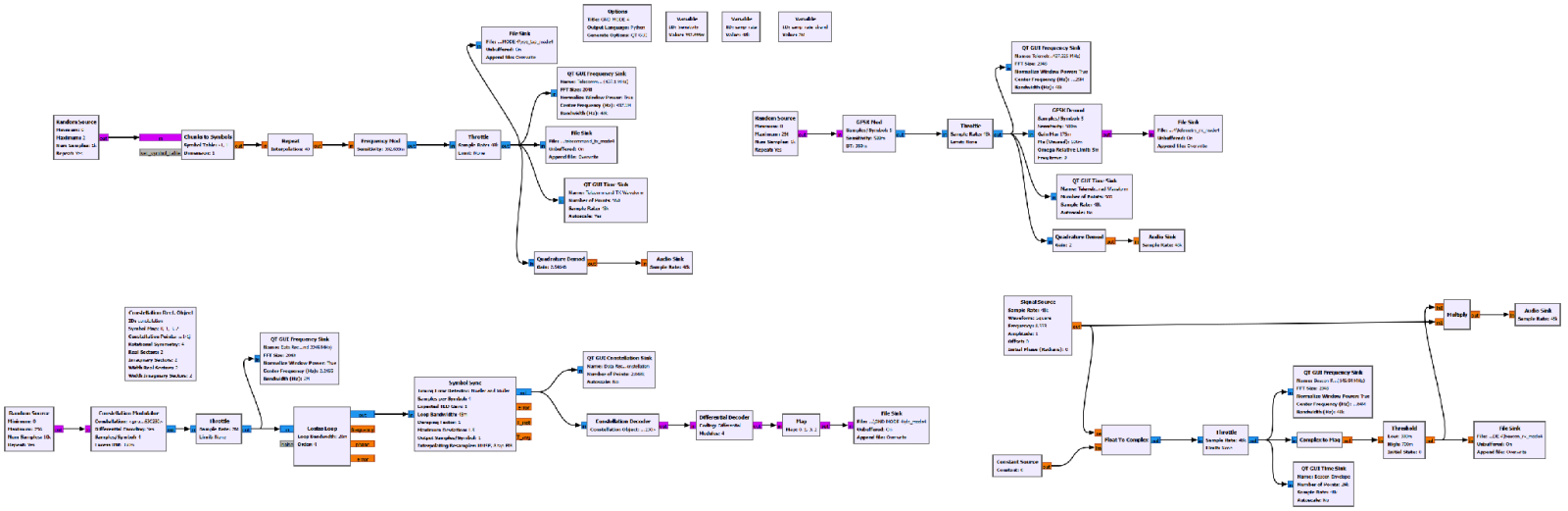


Fig. 4.6 Ground station (GND) flowgraph canvas.

Validation note. In the offline pipeline the dedicated complex signal tap intended for the S-band eye and constellation figures is connected to the telecommand baseband rather than to the S-band symbol synchronizer output, so a measured offline constellation and eye diagram are not available. The decoded symbol statistics and the live runtime constellation both confirm correct data recovery, and the offline constellation and eye figures can be produced after the tap is reconnected to the S-band chain.

GNU Radio Runtime Plots (GND)

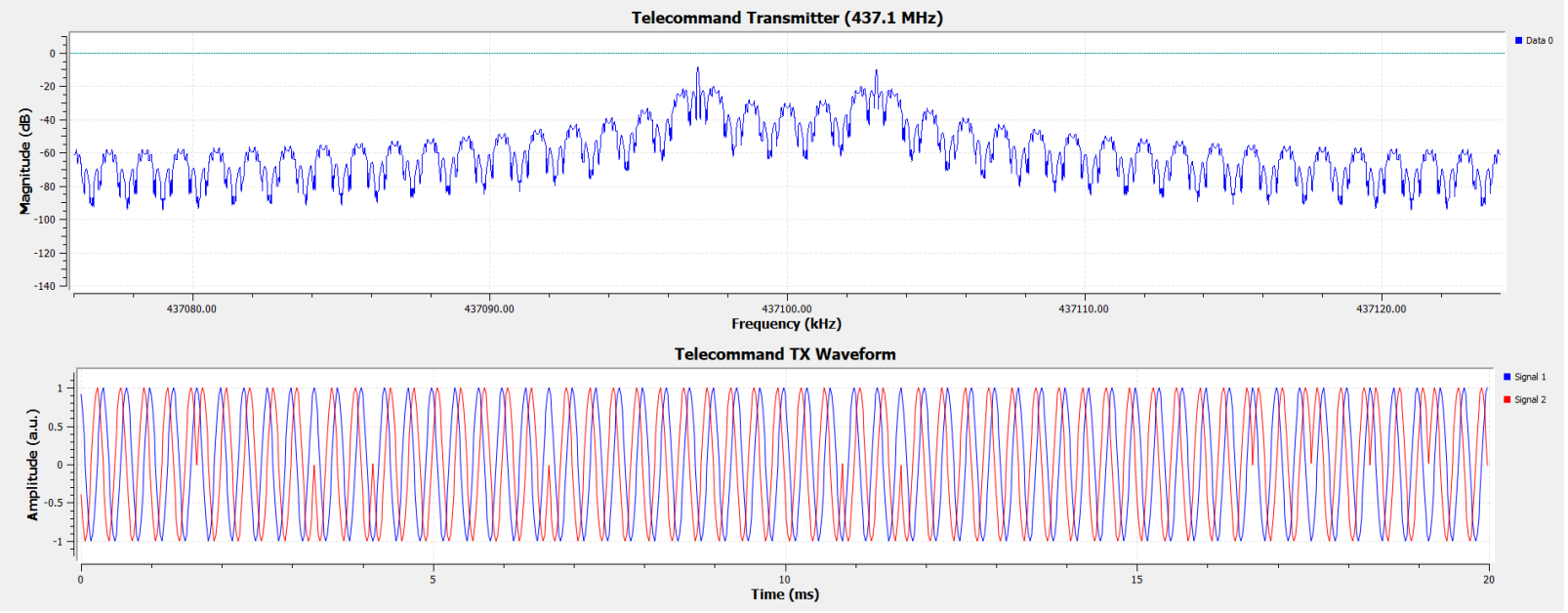


Fig. 4.7 (GND) Telecommand transmitter spectrum and TX waveform (437.1 MHz BFSK)

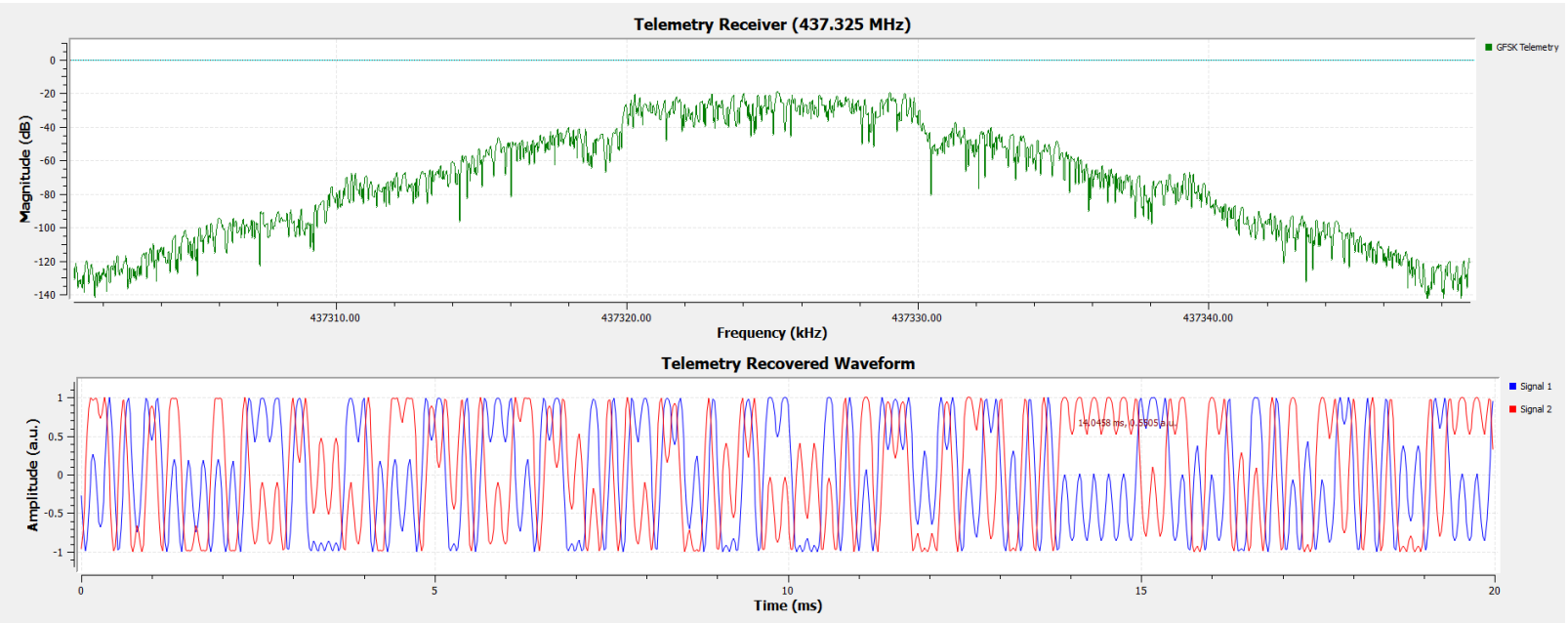


Fig. 4.8 (GND) Telemetry receiver spectrum and recovered waveform (437.325 MHz GFSK)

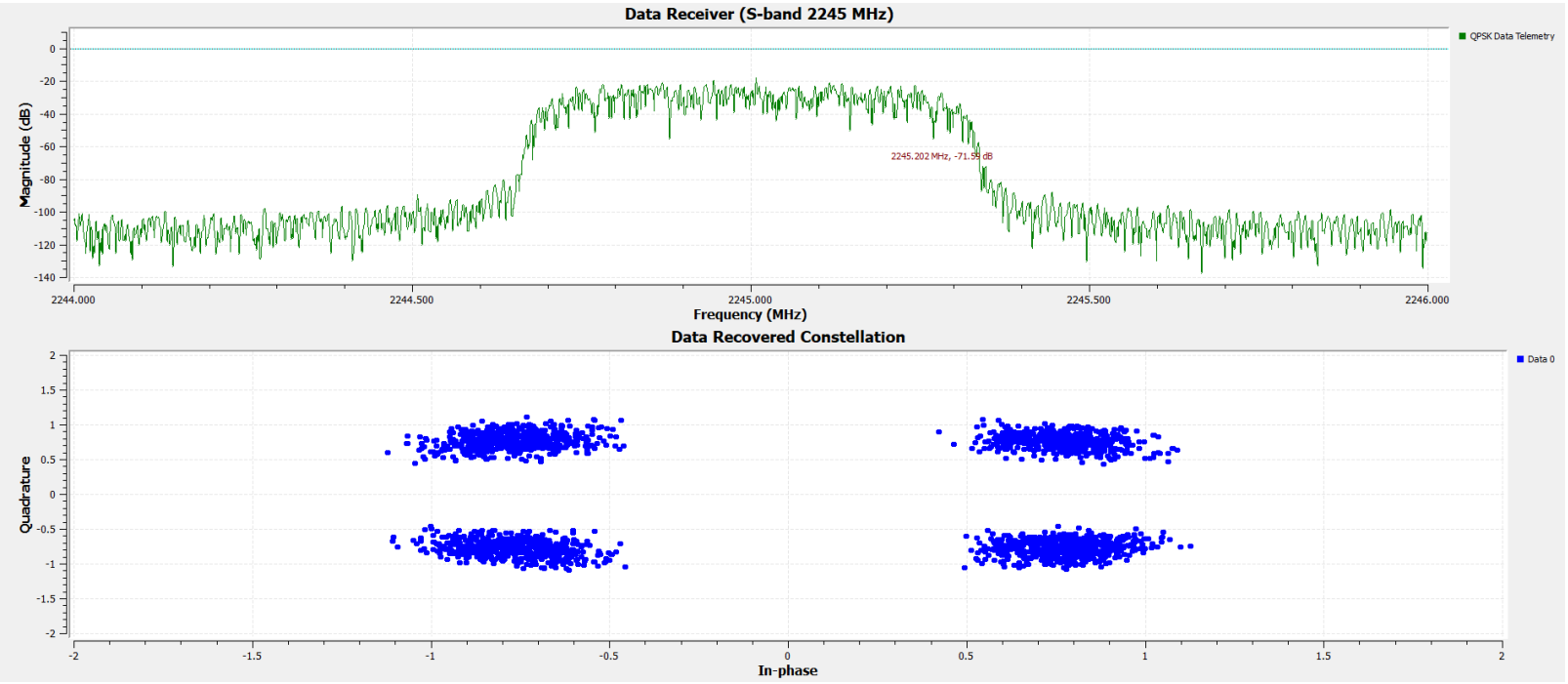


Fig. 4.9 (GND) Data receiver S-band spectrum and recovered QPSK constellation (2245 MHz)

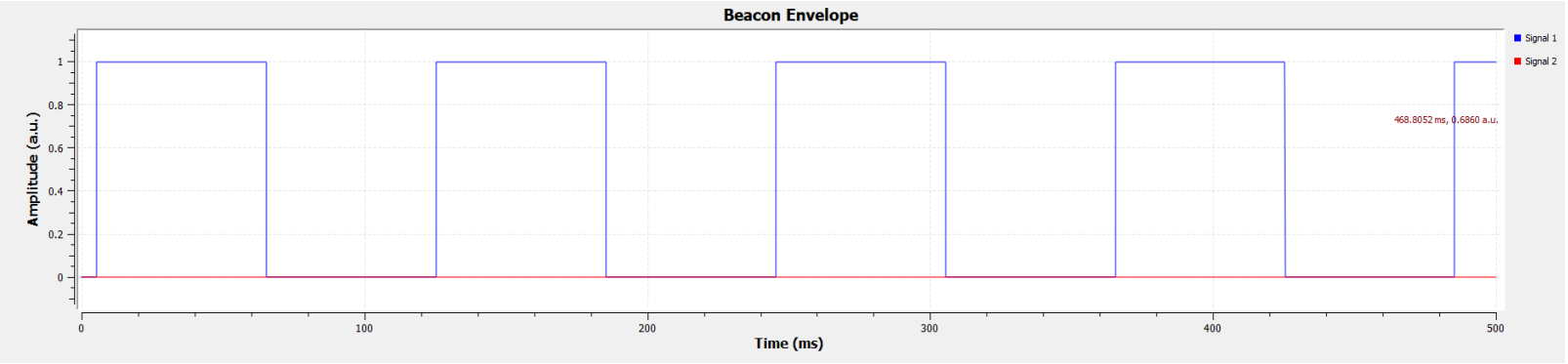


Fig. 4.10 (GND) Beacon envelope, recovered on-off keying (145.84 MHz)

Recovered Signal Figures (offline analysis)

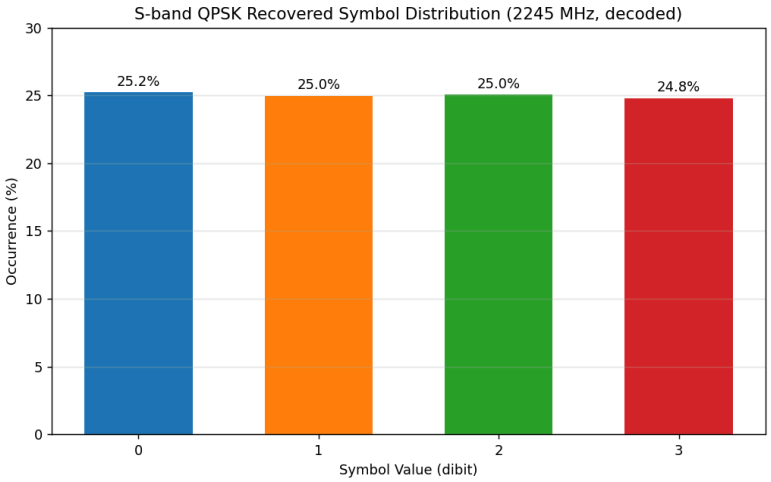


Fig. 4.11 S-band QPSK recovered symbol distribution (2245 MHz)

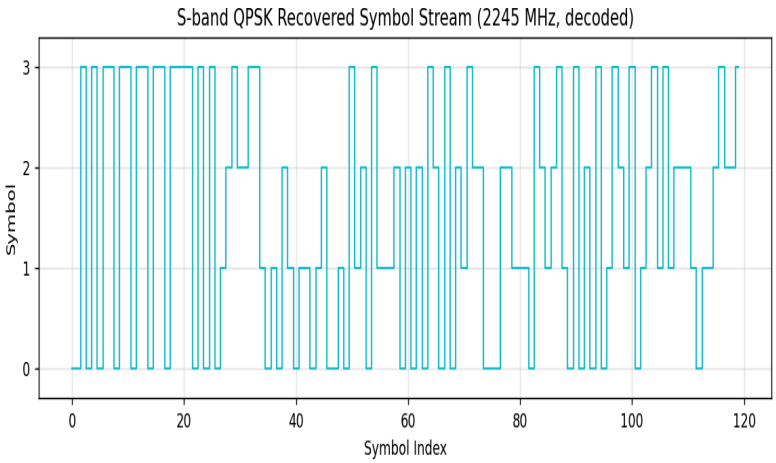


Fig. 4.12 S-band QPSK recovered symbol stream (2245 MHz)

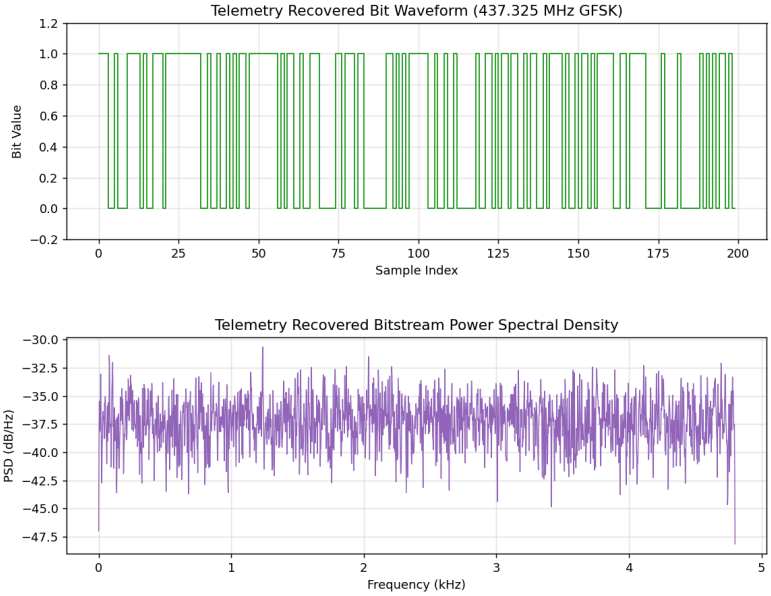


Fig. 4.13 Telemetry RX bit waveform and PSD (437.325 MHz GFSK)

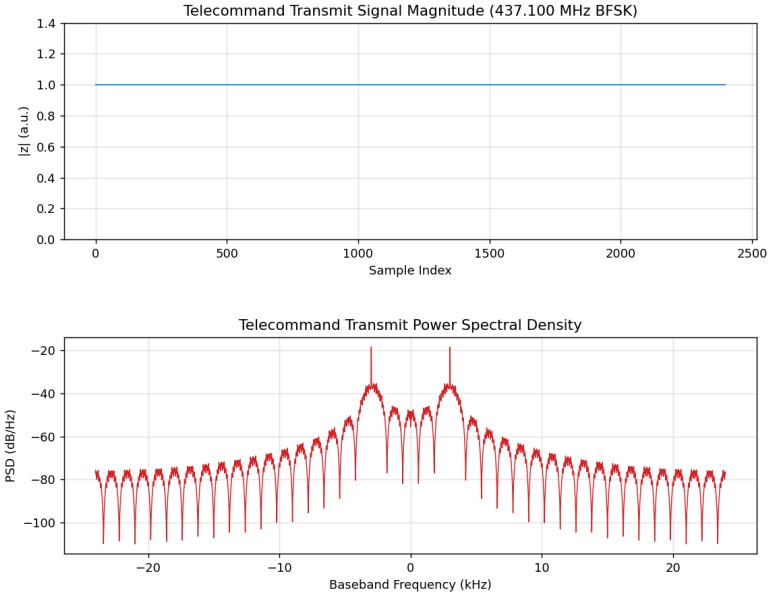


Fig. 4.14: Telecommand TX magnitude and PSD (437.100 MHz BFSK)

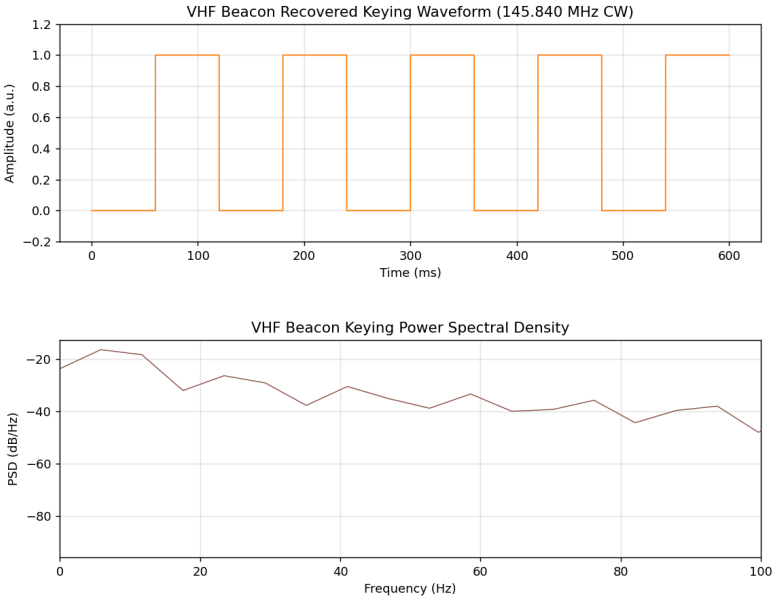


Fig. 4.15 VHF beacon keying waveform and PSD (145.840 MHz CW)

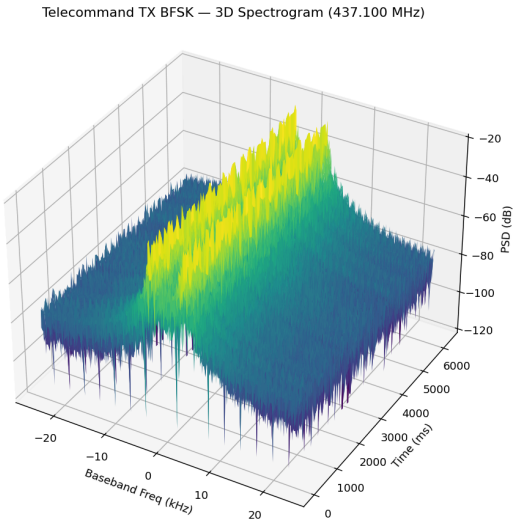


Fig. 4.16 Telecommand BFSK 3D spectrogram

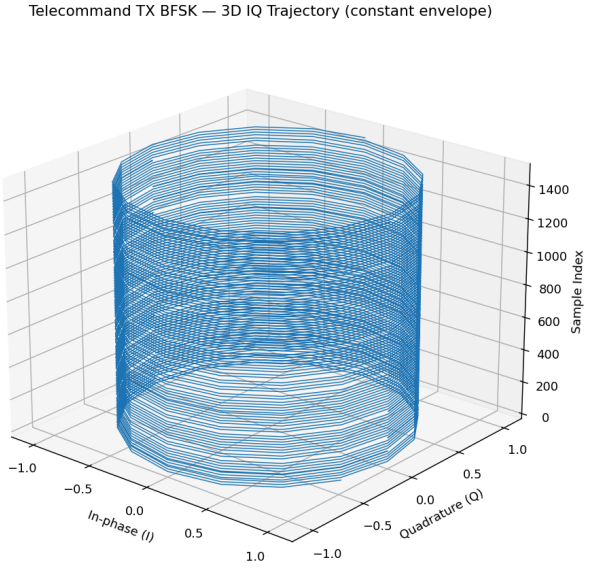


Fig. 4.17: Telecommand BFSK 3D IQ trajectory

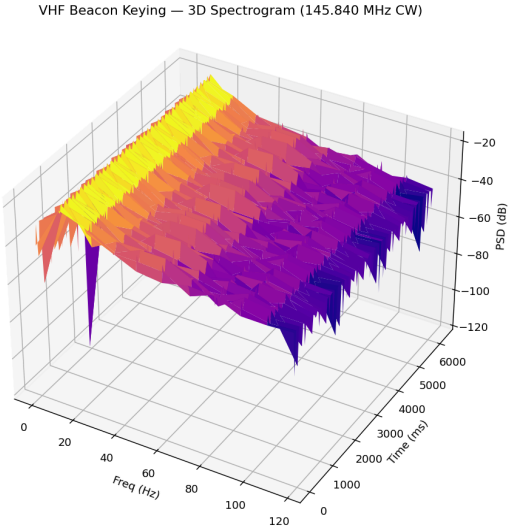


Fig. 4.18: VHF beacon keying 3D spectrogram

Four-View Telecommand Spectrogram

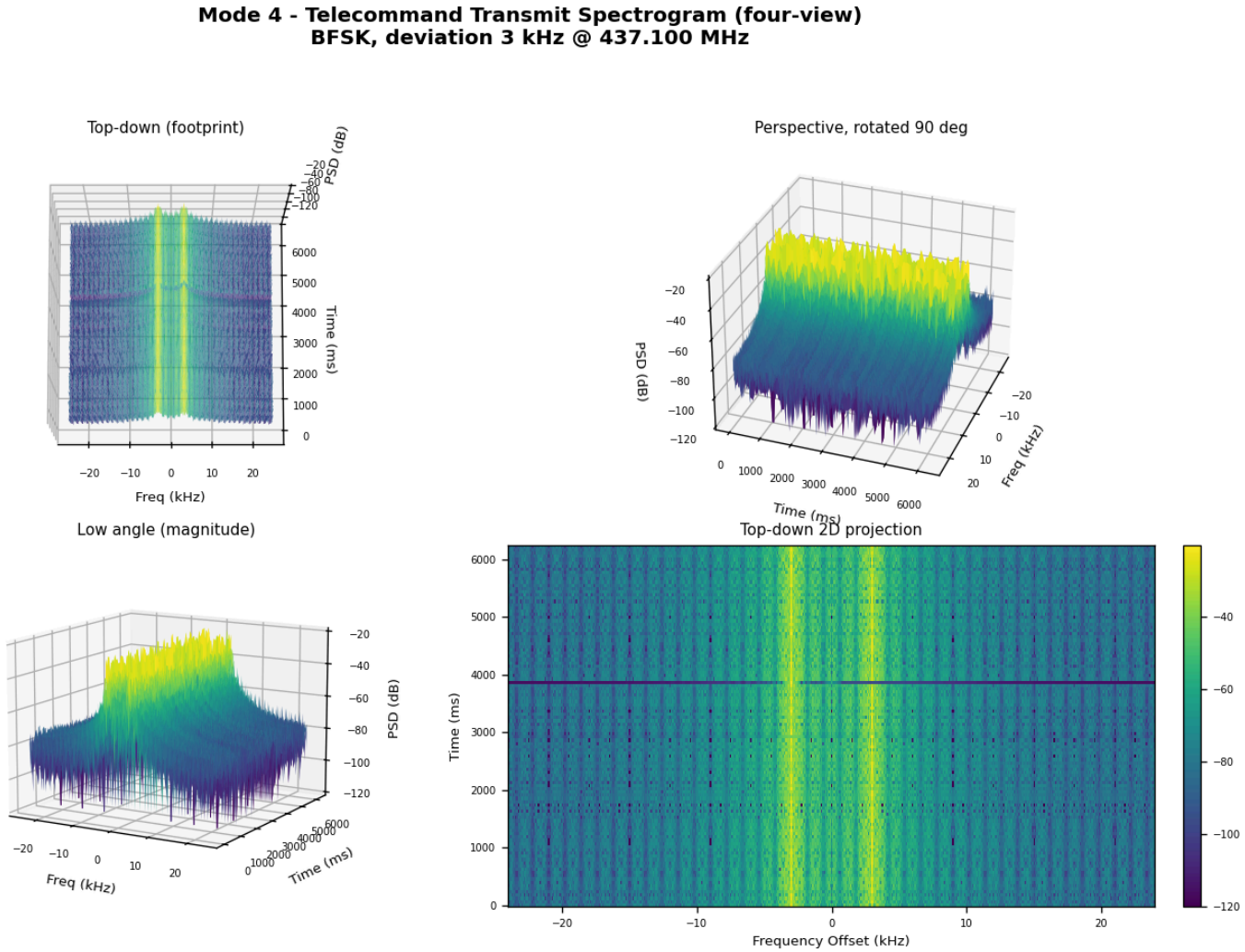


Fig. 4.19. Telecommand transmit spectrogram shown from four perspectives (top-down, rotated, low-angle, and 2D projection).

4.5 Mode 5: Emergency / VHF

Satellite Terminal (STL)

Mode 5 is the emergency configuration. The telecommand uplink remains BFSK at 437.100 MHz, but the telemetry downlink is moved to VHF GFSK at 145.950 MHz to provide a robust low-frequency return path. Mode 5 carries no payload data channel.

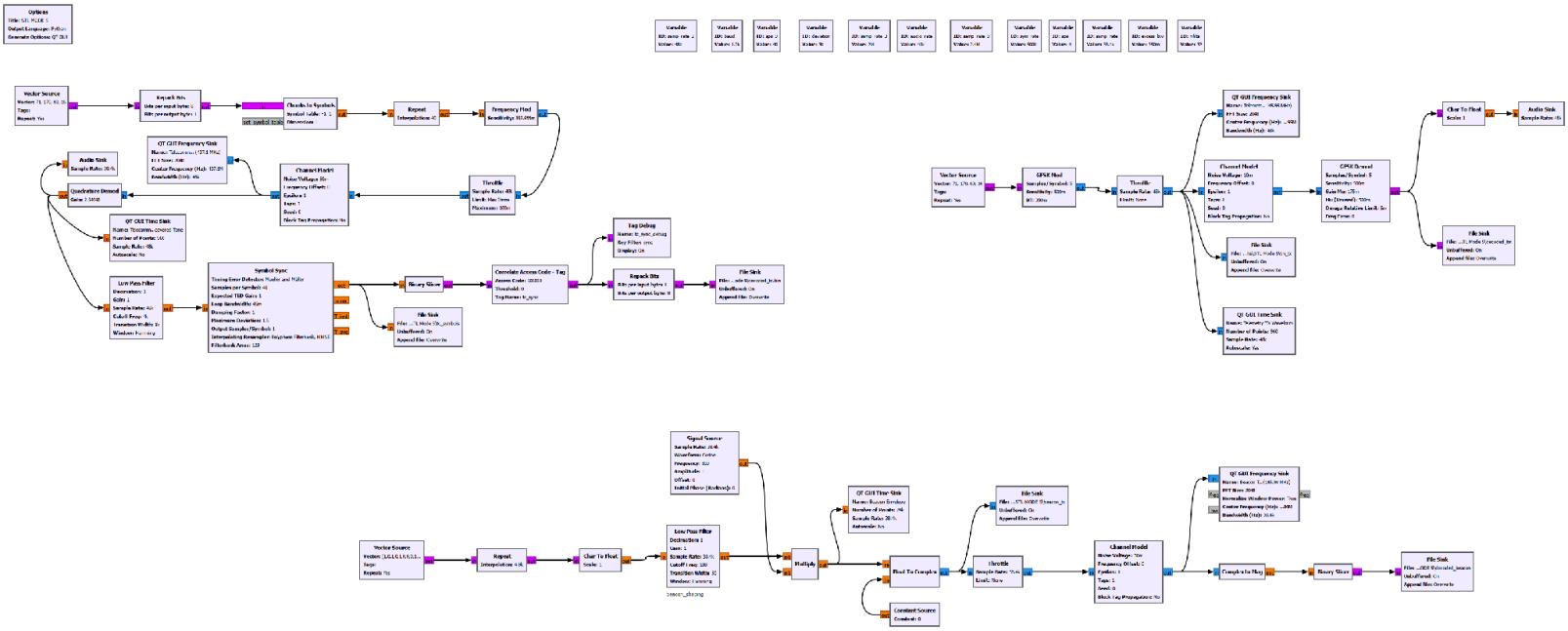


Fig. 5.1 Satellite (STL) flowgraph canvas.

GNU Radio Runtime Plots (STL)

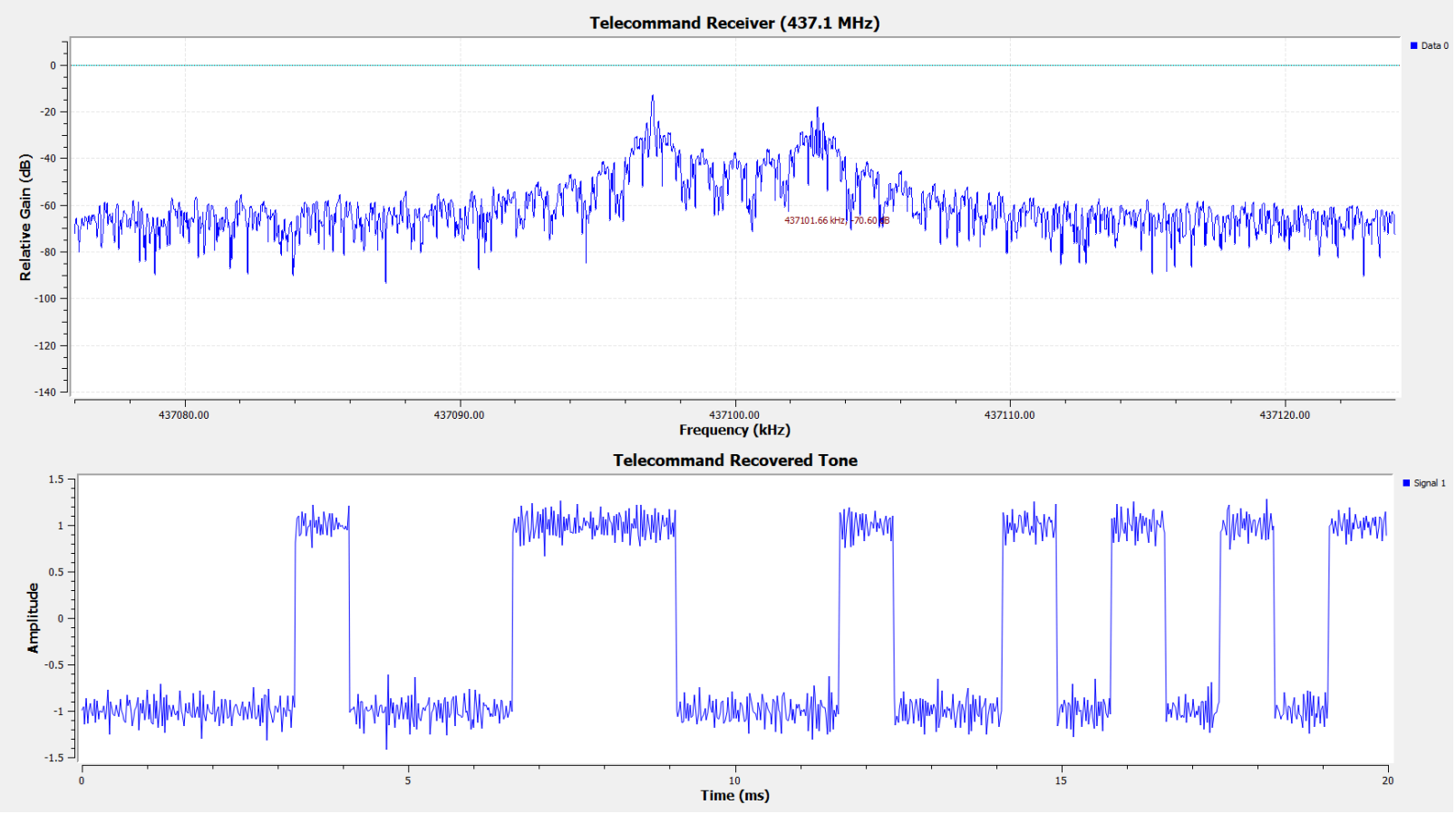


Fig. 5.2 (STL) Telecommand receiver spectrum and recovered tone (437.1 MHz BFSK)

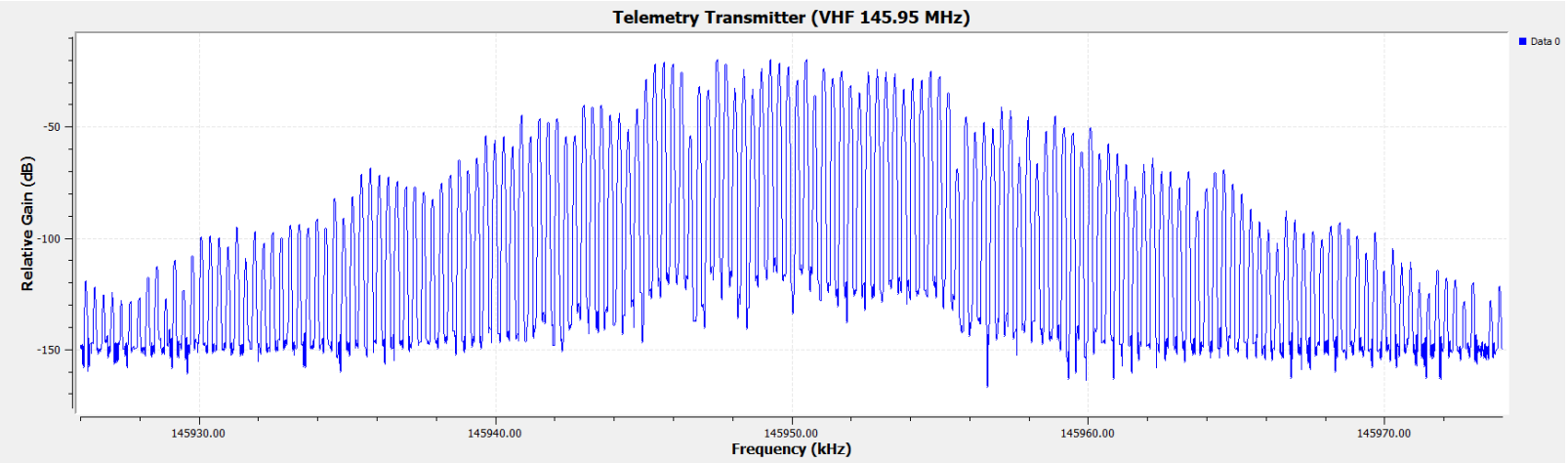


Fig. 5.3 (STL) Telemetry transmitter spectrum (VHF 145.95 MHz GFSK)

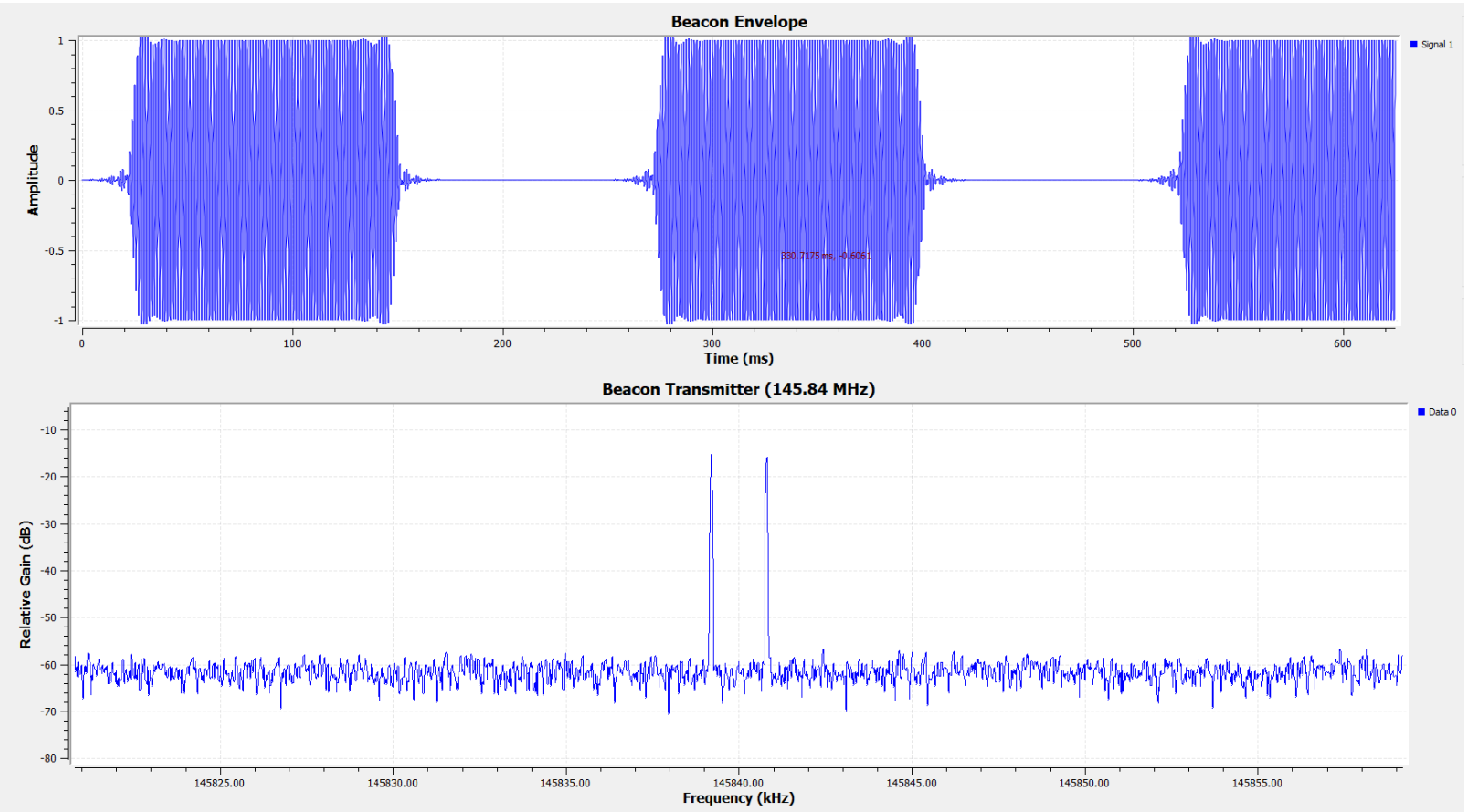


Fig. 5.4 (STL) Beacon envelope and transmitter spectrum (145.84 MHz CW)

Ground Station (GND)

The Mode 5 ground station transmits the BFSK command uplink and receives the VHF telemetry downlink at 145.950 MHz, which demodulates to a balanced bit stream. The command and telemetry links are healthy, and the file sinks route correctly to the Mode 5 output directory.

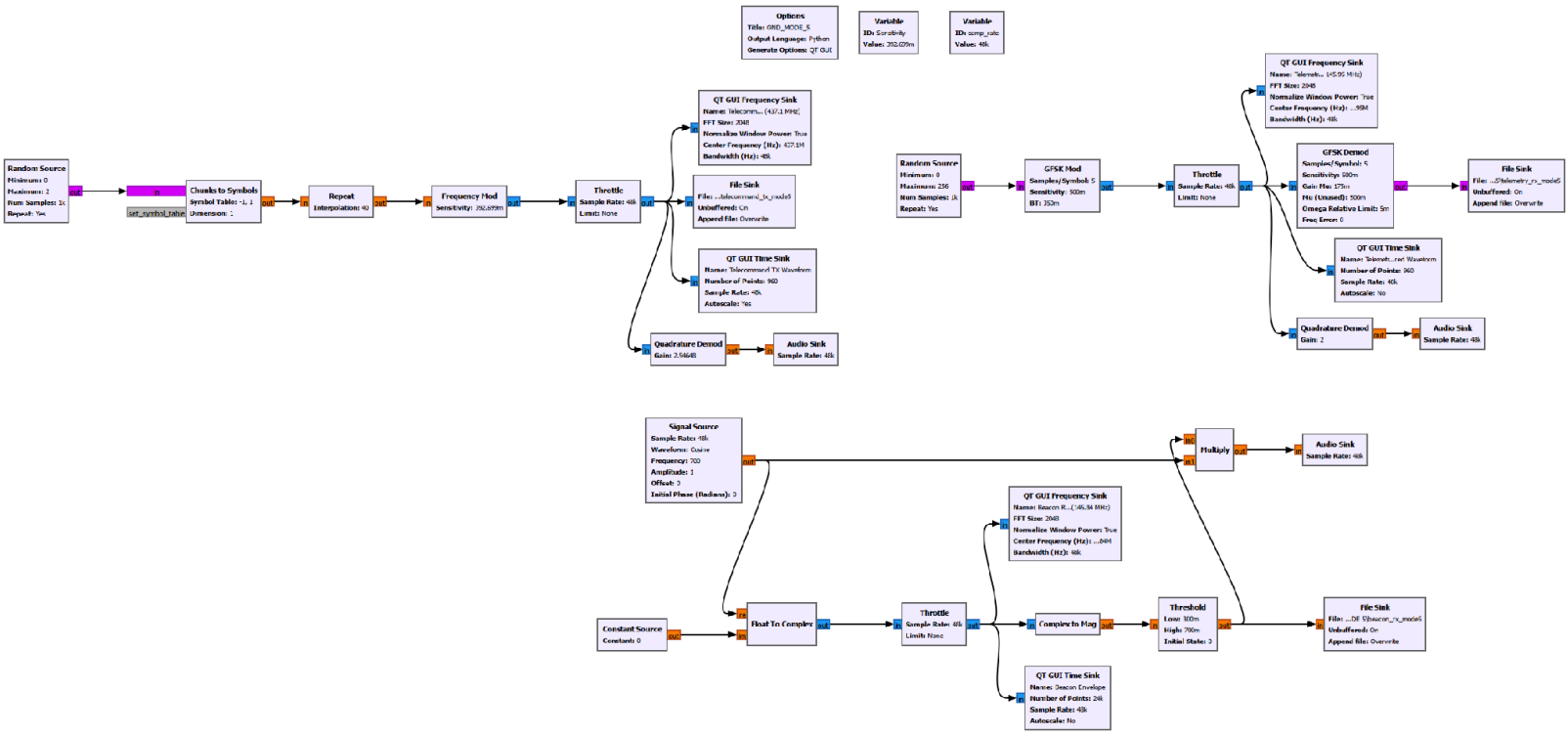


Fig. 5.5 Ground station (GND) flowgraph canvas.

Validation note. The ground canvas retains a beacon detector chain from the earlier modes, but its test source is a 700 Hz cosine rather than the 8.333 Hz keying square wave used in Modes 2 through 4. The magnitude detector rectifies this cosine to a 1400 Hz tone, so the recovered beacon envelope is a continuous tone rather than continuous-wave keying, as seen in the ground beacon runtime plot. This chain is a residual artifact, and its output is included only as a validation figure.

GNU Radio Runtime Plots (GND)

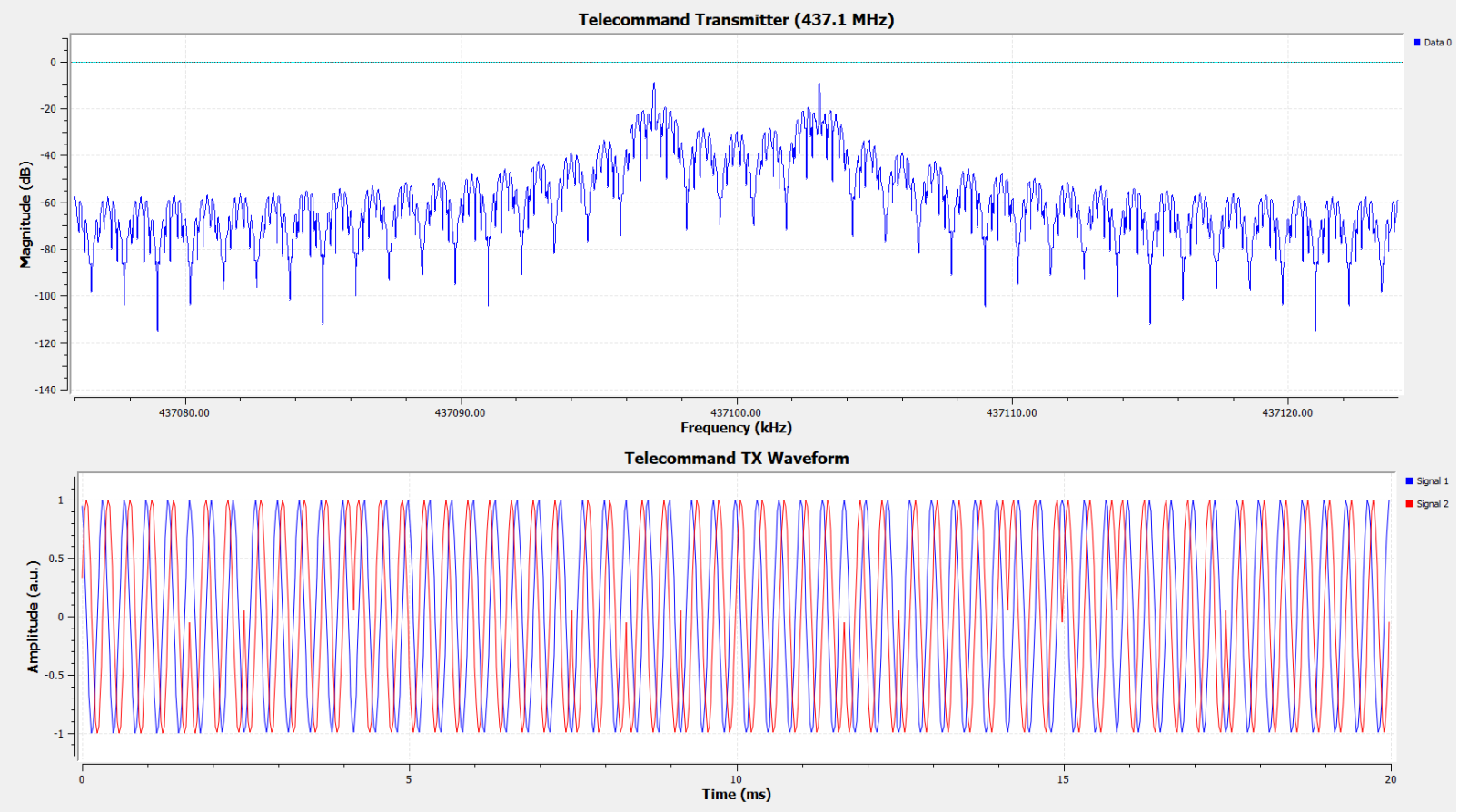


Fig. 5.6 (GND) Telecommand transmitter spectrum and TX waveform (437.1 MHz BFSK)

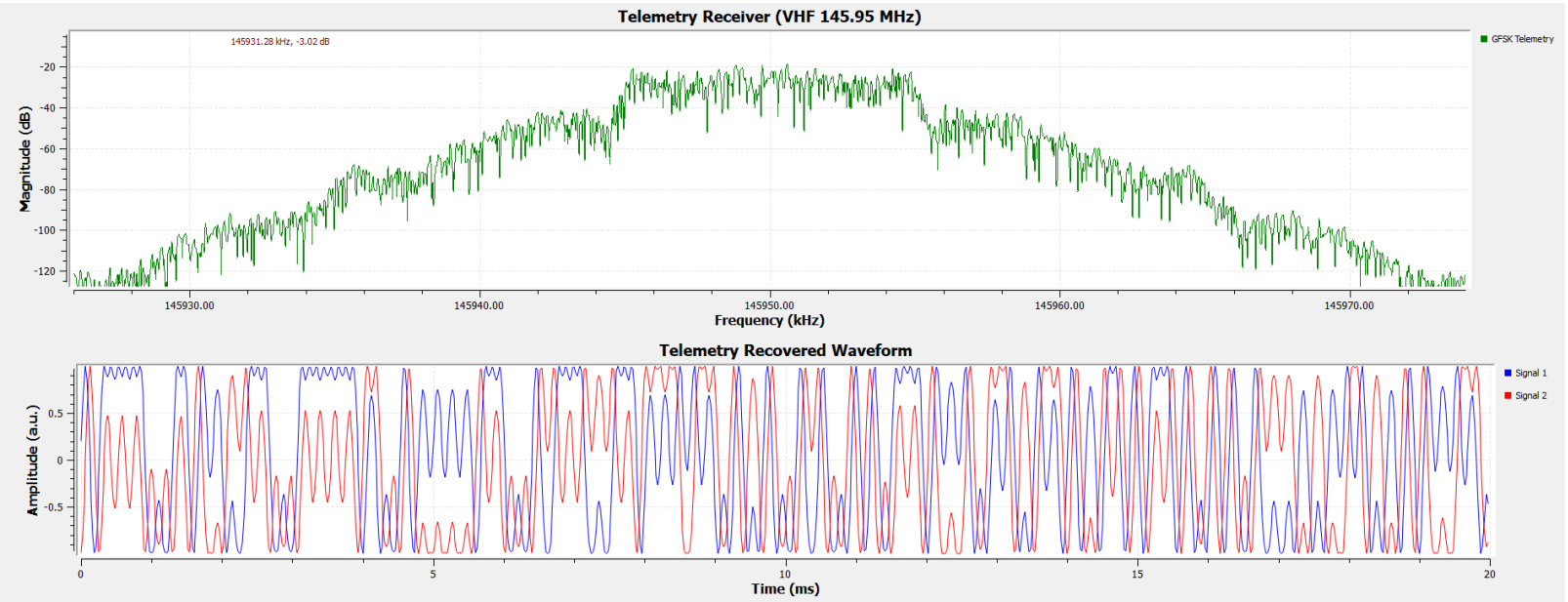


Fig. 5.7 (GND) Telemetry receiver spectrum and recovered waveform (VHF 145.95 MHz GFSK)

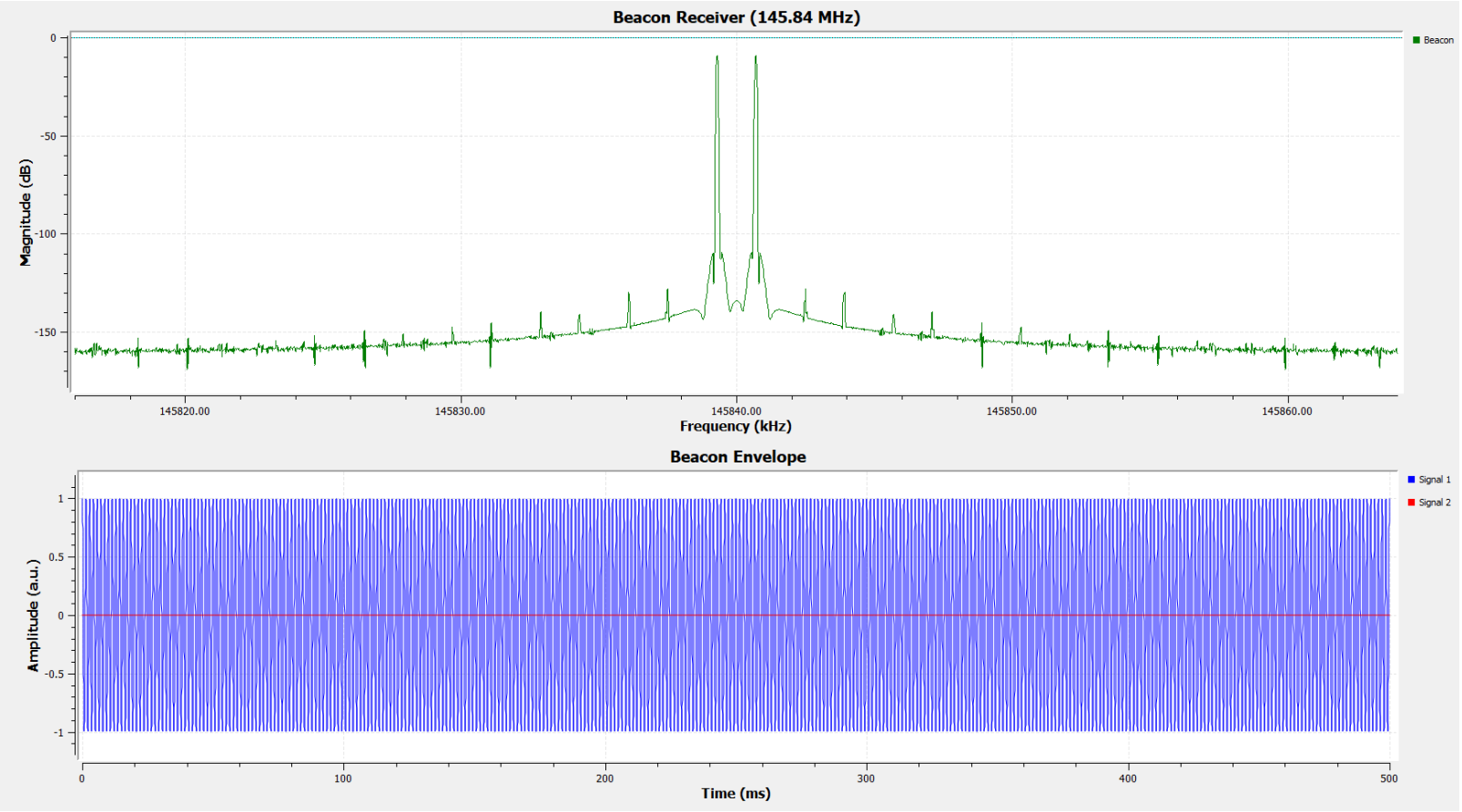
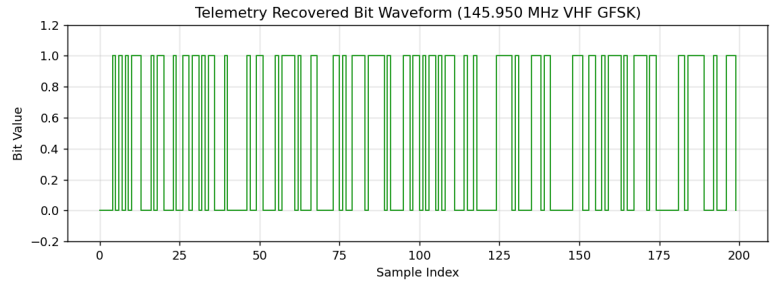


Fig. 5.8 (GND) Beacon receiver spectrum and envelope, 700 Hz cosine test-tone artifact (see note)

Recovered Signal Figures (offline analysis)



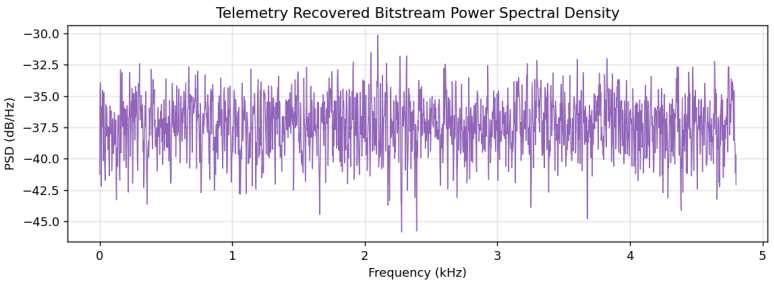
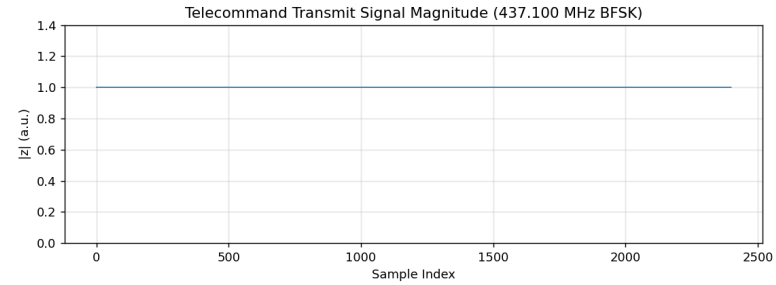


Fig. 5.9 Telemetry RX bit waveform and PSD (145.950 MHz VHF GFSK)



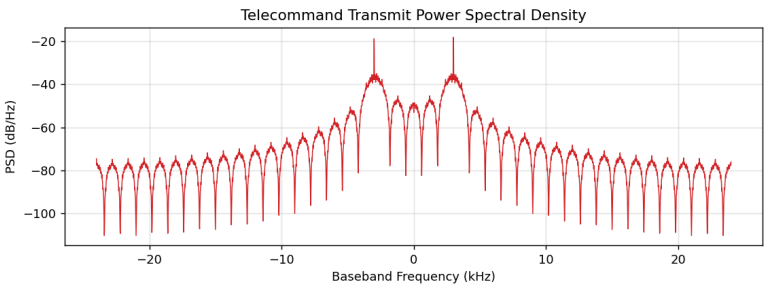


Fig. 5.10: Telecommand TX magnitude and PSD (437.100 MHz BFSK)

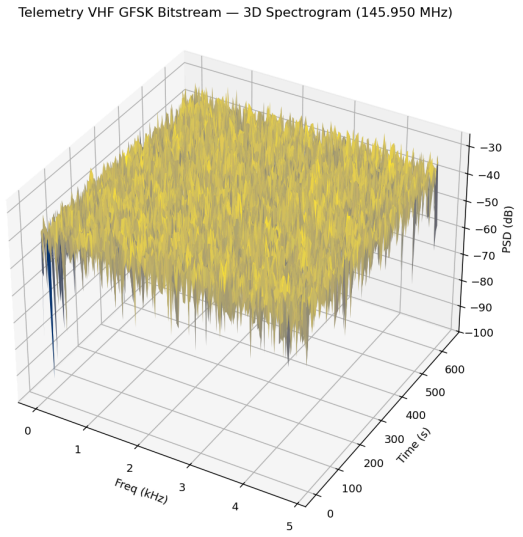


Fig. 5.11 Telemetry VHF GFSK bitstream 3D spectrogram (145.950 MHz)

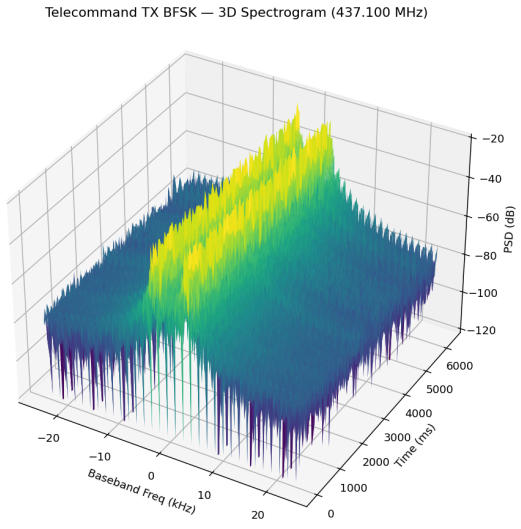


Fig. 5.12 Telecommand BFSK 3D spectrogram

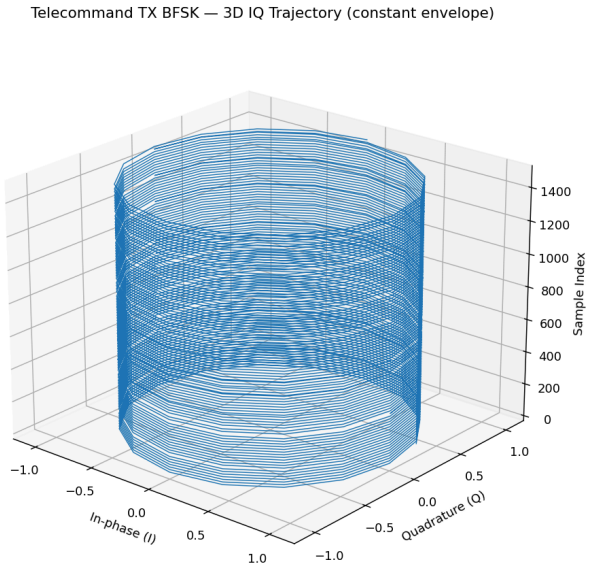


Fig. 5.13: Telecommand BFSK 3D IQ trajectory

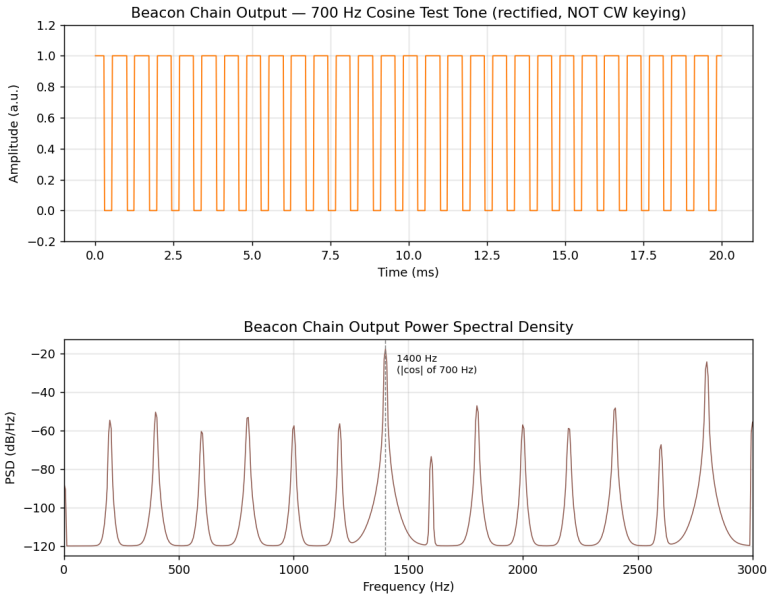


Fig. 5.14 Beacon chain output, 700 Hz cosine test tone (validation artifact)

Four-View Telecommand Spectrogram

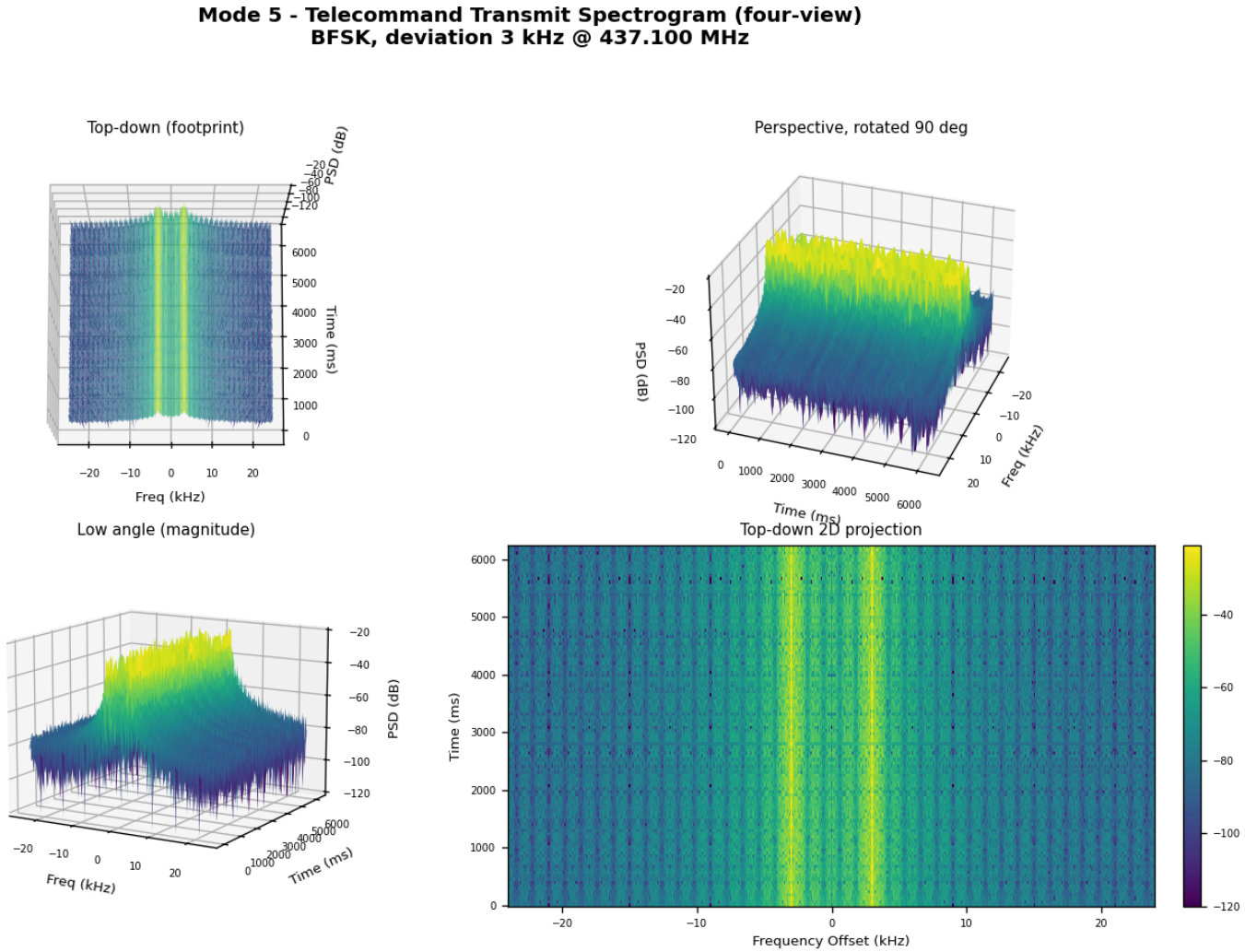


Fig. 5.15: Telecommand transmit spectrogram shown from four perspectives (top-down, rotated, low-angle, and 2D projection).

5. Summary

This report presented the satellite and ground flowgraph configurations for all five operational modes of the CubeSat communications subsystem, together with the live GNU Radio runtime plots for every mode and the offline recovered signal figures for the inter-satellite, high data downlink, and emergency modes. Across the modes examined, the command, telemetry, and beacon links behave as designed, and the S-band data decode in the high data downlink mode recovers balanced QPSK symbols, confirmed by the live ground constellation.

Two implementation items were identified during validation and are noted in the relevant mode sections. In the high data downlink mode, the offline complex signal tap intended for the S-band constellation and eye figures is connected to the telecommand baseband rather than the S-band chain, which prevents the production of those particular offline figures until the tap is reconnected; the live runtime constellation is unaffected. In the emergency mode, a residual beacon detector chain with a 700 Hz cosine test source remains in the ground canvas even though it does not represent continuous-wave keying. A minor label item was also noted, where the satellite mode 4 telemetry sink title carries a VHF tag on a UHF carrier. None of these items affect the command or telemetry link performance.